

A close-up, low-angle shot of solar panels, showing the grid lines and the reflective surface, with a warm, golden light reflecting off the panels.

INDIA'S ROLE IN DIVERSIFYING GLOBAL CLEAN ENERGY SUPPLY CHAINS

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India's Role in Diversifying Global Clean Energy Supply Chains

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PREFACE

BY SHAYAK SENGUPTA

To avert the worst effects of climate change, the energy transition must continue in a world that is becoming more geopolitically and geoeconomically fragmented.¹ Ongoing armed conflicts in Europe and the Middle East and a fractured trade relationship between the world's two largest economies, the United States and China, mean the energy transition cannot take economic integration and its accompanying benefits for granted.²

National governments are increasingly becoming comfortable with tying climate and energy policies to economic competition. They look to capture the economic opportunities associated with the transition by securing their shares of raw materials, technologies, and manufacturing capacities needed for clean energy.³

Carrot and stick efforts span both developed and developing regions. The United States passed the Inflation Reduction Act (IRA) to subsidize clean energy manufacturing and deployment while Europe has established a Carbon Border Adjustment Mechanism (CBAM) to penalize imported goods from countries with emissions intensive production.⁴ In India, Production-Linked Incentives (PLI) offer a suite of financial and non-financial incentives for companies to set up clean energy manufacturing in the country and impending tariffs on imported Chinese solar panels look to give a leg up to domestic manufacturers.⁵ Indonesia has banned the export of raw nickel (a key component of electric vehicle batteries), forcing producers set up mineral processing within the country.⁶

While this fragmentation offers economies the opportunity for a share of the economic pie from the energy transition, no single country will be able to manufacture all clean energy technologies and associated inputs.⁷ Financial and investment flows must accompany ambitions for clean energy manufacturing to make these opportunities a reality. Consequently, cooperation between countries to manage differences arising from fragmentation will be necessary for the energy transition.⁸

¹ Gita Gopinath, "How Policymakers Should Handle a Fragmenting World," *Foreign Policy*, February 6, 2024, <https://foreignpolicy.com/2024/02/06/how-policymakers-should-handle-a-fragmenting-world/>; Charlotte Gardes-Landolfini, Pierpaolo Grippa, William Oman, and Sha Yu, "Staff Climate Notes: Energy Transition and Geoeconomic Fragmentation," International Monetary Fund, November 28, 2023, 1, <https://www.imf.org/en/Publications/staff-climate-notes/Issues/2023/11/16/Energy-Transition-and-Geoeconomic-Fragmentation-Implications-for-Climate-Scenario-Design-541097>.

² Shekhar Aiyar et al., "Geo-Economic Fragmentation and the Future of Multilateralism," Staff Discussion Notes International Monetary Fund, January 15, 2023, 1, <https://www.elibrary.imf.org/view/journals/006/2023/001/article-A001-en.xml>.

³ Bentley Allan, Joanna I. Lewis, and Thomas Oatley, "Green Industrial Policy and the Global Transformation of Climate Politics," *Global Environmental Politics*, November 28, 2021, 1–19, https://doi.org/10.1162/glep_a_00640.

⁴ The White House, "Building a Clean Energy Economy: A Guidebook to the Inflation Reduction Act's Investments in Clean Energy and Climate Action," January 9, 2023, <https://www.whitehouse.gov/wp-content/uploads/2022/12/Inflation-Reduction-Act-Guidebook.pdf>; European Commission, "Carbon Border Adjustment Mechanism," European Commission, 2024, https://taxation-customs.ec.europa.eu/carbon-border-adjustment-mechanism_en.

⁵ Rajesh Kumar Singh, "India Plans Stricter Local Material Norm for Solar Module Makers," *Bloomberg*, October 20, 2023, <https://www.bloomberg.com/news/articles/2023-10-20/india-plans-stricter-local-material-norm-for-solar-module-makers>.

⁶ International Energy Agency, "Prohibition of the Export of Nickel Ore," IEA, December 12, 2023, <https://www.iea.org/policies/16084-prohibition-of-the-export-of-nickel-ore>.

⁷ Monika Mondal and Mitul Kajaria, "India Wants to Build a Solar Manufacturing Ecosystem to Rival China," *Climate Home News*, January 30, 2024, <https://cleanenergyfrontier.climatechangenews.com/india-wants-own-solar-industry-break-reliance-china/>.

⁸ Aiyar et al., "Geo-Economic Fragmentation and the Future of Multilateralism,"; Gardes-Landolfini, "Energy Transition and Geoeconomic Fragmentation."

Given this context, India's G20 presidency in 2023 elevated several topics to prioritize clean energy technology, manufacturing, and finance between the 20 largest economies in the world.⁹ India plays a crucial role in this grouping. It ranks among the largest economies, greenhouse gas emitters, and consumers of energy. As the largest source of global energy demand in the coming years, India's own clean energy transition and economic development will influence whether the world succeeds in meeting its climate goals.¹⁰

This volume serves two objectives. First, it aspires to contextualize India in the emerging global landscape of manufacturing and supply chains of three clean energy sectors: hydrogen, electric batteries, and solar. Second, it seeks to enhance understanding of how these contexts paired with relevant priorities from India's 2023 G20 presidency can inform global clean technology and finance partnerships centering India. The volume aims to give a global policy audience a better understanding of where and how India could play a role in the supply of these emerging, transformative clean energy sectors.

In the first piece, Raj Sawhney analyzes the costs of production of renewable-based hydrogen (green hydrogen) in India. Green hydrogen could play a role to decarbonize industrial sectors, but India faces competition from many G20 countries setting strategies to capture parts of a growing, nascent green hydrogen market. Higher capital investments mean lower-cost finance will be crucial to making India's green hydrogen production cost-competitive.

In the second and third pieces, Gregory Wischer looks at India's role in global supply chains of electric vehicle batteries and solar panels, respectively. China currently dominates supply chains of both sectors. While India, like the United States and Europe, lacks reserves of critical, raw materials that go into electric vehicles like lithium, nickel, and cobalt, it does produce sizable amounts of ancillary inputs like graphite and manganese. Consequently, international partnerships could fuel growth of India's share in parts of battery manufacturing, to meet both domestic and export demand. Likewise in solar, India, along with the United States and Europe, are looking to grow domestic shares of solar manufacturing. While India already manufactures (and exports) solar panels and cells, it still depends on China for upstream inputs. Costs of energy and financing must decrease in India to grow the country's nascent shares of raw materials for solar manufacturing.

Lastly, in the fourth piece, I look at priorities in clean energy technology and finance from India's G20 Presidency in 2023 and how they square with the country's role in global hydrogen, solar, and battery supply chains. Among geopolitical and geo-economic fragmentation, finance for the energy transition must address emerging markets' desire to capture additional value from clean energy manufacturing and production. India is no exception to this trend, but the country offers a unique position as a clean energy technology supplier for both emerging and developed markets.

⁹ Indian G20 Presidency, "G20 New Delhi Leaders' Declaration," September 10, 2023, <https://www.mea.gov.in/Images/CPV/G20-New-Delhi-Leaders-Declaration.pdf>; Chietigij Bajpae, "The G20 Showcases India's Growing Power. It Could Also Expose the Limits of Its Foreign Policy," September 7, 2023, <https://www.chathamhouse.org/2023/09/g20-showcases-indias-growing-power-it-could-also-expose-limits-its-foreign-policy>; Takeo Konishi, "India's G20 Presidency: An Opportunity to Steer the World Toward Inclusive and Sustainable Growth," Asian Development Bank, June 19, 2023, <https://www.adb.org/news/features/indias-g20-presidency-opportunity>.

¹⁰ International Energy Agency, "India Energy Outlook 2021," World Energy Outlook Special Report, 2021, <https://www.iea.org/reports/india-energy-outlook-2021>.

DECODING INDIA'S GREEN HYDROGEN POTENTIAL

BY RAJ SAWHNEY

SUMMARY

India's position in the global green hydrogen market is complex, marked by both challenges and opportunities. Currently, green hydrogen production costs in India are estimated between \$5.30 to \$6.70 per kilogram (kg), approximately 40 percent more expensive than the cheapest G20 green hydrogen producer, Australia. To compete as a hydrogen exporter, India must reduce its production costs by around 30 – 40 percent. Furthermore, a domestic transition from traditional natural-gas derived (grey) hydrogen to renewable energy-derived (green) hydrogen will incur a substantial green premium, approximately 2 – 3.5 times current grey hydrogen prices. The bulk of green hydrogen production costs, nearly 95 percent, are driven by capital costs, like equipment to produce hydrogen and renewable energy, and the cost of capital. Thus, investment and policy initiatives targeted towards achieving green hydrogen cost parity should focus on these elements. India's large fertilizer and refinery industries represent a pathway to reduce capital costs by offering a tangible demand baseline, an element that, when missing, could significantly increase the cost of capital. This demand baseline represents a significant advantage, as only 10 percent of planned green hydrogen production by 2030 have secured off-takers.¹ As currently constructed, India may need more than the \$2.1 billion it has already committed to make green hydrogen a viable component of its energy strategy.

1. INTRODUCTION

India is undertaking a rapid clean energy transition. With a promise to reach net-zero emissions by 2070, India must decarbonize an array of sectors including transportation, heavy industry, and electricity generation. Traditional renewable energy sources like wind and solar are instrumental in tackling many of these challenges. India has already installed 180 gigawatts (GW) of renewable energy capacity and aims to achieve 500 GW by 2030.² However, hard-to-abate industries such as fertilizer production, refineries, steel manufacturing, and chemical production remain difficult to decarbonize and are crucial to not only meeting the country's economic aspirations, but also as a source of new future emissions.³

The element hydrogen could aid in decarbonizing these sectors, but 96 percent of the world's hydrogen is generated through emissions-intensive processes due to lower costs than emissions-free production. As global decarbonization efforts grow, many countries, including India, are looking to reduce costs of cleaner, emission-free hydrogen ("clean hydrogen"). These efforts not only seek to use the element in an energy transition, but also as an economic opportunity, by capitalizing on the emerging hydrogen value chain that may transform its use from predominately local to regional and global.⁴

Yet, India's role in the future global hydrogen economy is uncertain. The nascent hydrogen value chain is complex, and countries are determining how they will integrate within this growing market. Among the largest economies in the world, the G20, most

¹ "Hydrogen offtake is tiny but growing," *BNEF*, November 14, 2023, <https://about.bnef.com/blog/hydrogen-offtake-is-tiny-but-growing/#:~:text=Many%20companies%20have%20plans%20to,for%20clean%20hydrogen%20and%20derivatives>.

² Saurabh Handa, "India Clean Energy: Progress and Policy," Citi Group, October 31, 2023, <https://www.citigroup.com/global/insights/global-insights/india-clean-energy-progress-policy>.

³ Simon Bennett, "India Energy Outlook 2021," IEA, February 2021, <https://www.iea.org/reports/india-energy-outlook-2021>.

⁴ Timur Gül et al., "Global Hydrogen Review 2023," IEA, September 2023, <https://www.iea.org/reports/global-hydrogen-review-2023>.

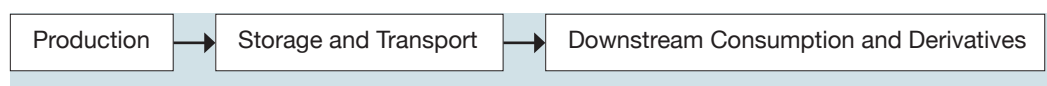
countries have stated intentions to become net exporters of hydrogen (see Table 2).

In the case of India, and many other stated exporters, the feasibility of these hydrogen export goals are unclear. Some previous analyses classify India as a clear (future) importer or bystander of clean hydrogen and many of its derivatives (such as ammonia or methanol) while others conclude that India is posed to be a medium to large exporter in future hydrogen markets.^{5 6 7 8}

This analysis aims to build on this discussion by precisely quantifying India's position within the global hydrogen landscape through clean hydrogen production costs. The opening section defines the hydrogen value chain at-large and various hydrogen production methods. Next, the following section describes the current state of this value chain globally and domestically in India and then defines a modeling framework to estimate current and future production costs of clean hydrogen and their drivers within India with relevant comparisons traditional hydrogen production methods. Subsequently, the analysis contextualizes India's clean hydrogen production costs to the G20 at-large. The author then offers conclusions and recommendations to those looking to understand what role India will play in the future of global hydrogen.

2. DEFINING CLEAN HYDROGEN AND THE HYDROGEN VALUE CHAIN

Figure 1: The Green Hydrogen Value Chain



Source: Author adaptation from McKinsey analysis.⁹

The hydrogen value chain consists of three interconnected components: production, storage and transportation, and downstream use cases.¹⁰ After production, hydrogen can be stored in both liquid and gaseous states, or as chemical derivatives like ammonia. Pipelines, liquid hydrogen, and ammonia shipping are the primary transportation methods. Significant uncertainty exists within this stage specifically concerning the safety, cost, and accessibility of these storage and transport methods. While there are several hydrogen pipelines integrated globally, hydrogen shipping is largely untested, with the world's first liquified hydrogen carrier being built in 2020 and the first test shipment occurring in February 2022.¹¹ To date, no other carrier has been built or operationalized. Finally, in downstream use cases, hydrogen and its derivatives are the

⁵ Gül et al., "Global Hydrogen Review 2023."

⁶ Laima Eicke and Nicola De Blasio, "The Future of Green Hydrogen Value Chains," Belfer Center, October 2022, https://www.belfercenter.org/sites/default/files/files/publication/Paper_MappingHydrogen_Final.pdf.

⁷ Lucie Togni and Rami Fakhoury, "Regional Insights into Low-Carbon Hydrogen Scale Up," World Energy Council, April 2022, <https://www.worldenergy.org/publications/entry/regional-insights-low-carbon-hydrogen-scale-up-world-energy-council>.

⁸ Herib Blanco and Emanuele Taibi, "Global Hydrogen Trade to Meet the 1.5°C Climate Goal," IRENA, July 2022, <https://www.irena.org/publications/2022/Jul/Global-Hydrogen-Trade-Outlook>.

⁹ Arnout de Pee et al., "The clean hydrogen opportunity for hydrocarbon-rich countries," McKinsey, November 23, 2022, <https://www.mckinsey.com/industries/oil-and-gas/our-insights/the-clean-hydrogen-opportunity-for-hydrocarbon-rich-countries>.

¹⁰ de Pee et al., "The clean hydrogen opportunity for hydrocarbon-rich countries."

¹¹ Yukichi Takaoka, Kentarou Mizumukai, and Yuichi Kamenno, "Suiso Frontier, the First LH2 Carrier — Demonstration of Technologies and Operational Phase," June 2023, <https://onepetro.org/ISOPEIOPEC/proceedings-abstract/ISOPE23/All-ISOPE23/ISOPE-I-23-490/524383>.

primary inputs in several hard-to-abate industries such as fertilizers, refineries, and steel production.

Various hydrogen production methods and their associated greenhouse gas emissions are often classified by colors. The three most common production methods are grey, blue, and green hydrogen (Table 1). Grey hydrogen is hydrogen generated through natural gas, primarily through a process in which natural gas reacts with steam in the presence of a catalyst to produce hydrogen. Carbon dioxide is emitted as a byproduct of this process. The blue hydrogen synthesis process mirrors grey hydrogen, except for the addition of carbon capture technology, which captures a percentage of the CO₂ byproduct, reducing overall emissions. Green hydrogen is defined as hydrogen produced using purely renewable energy resources (like wind, solar, hydropower). Specifically, renewable energy powers an electrolyzer, where electricity splits a water molecule into its constituent hydrogen and oxygen molecules. This process is emissions-free so long as the input electricity is also emissions-free. Note that some countries, like India, also consider biomass-derived hydrogen to be in the “green” class.

This report focuses on this production step of this value chain, specifically quantifying and comparing India's green hydrogen production costs with the cost of other methods of hydrogen production.

Table 1: Hydrogen Colors Emissions Intensity and Price in India

	EMISSIONS INTENSITY (kg CO ₂ eq/kg H ₂)	PRICE (\$/kg)
BLUE HYDROGEN	5 – 8	\$2.70 – \$3.40
GREEN HYDROGEN	0	\$5.30 – \$6.70
GREY HYDROGEN	10 – 14	\$1.90 – \$2.40

Source: Author's analysis, with information from “Hydrogen offtake is tiny but growing,” BNEF, November 14, 2023, <https://about.bnef.com/blog/hydrogen-offtake-is-tiny-but-growing/#:~:text=Many%20companies%20have%20plans%20to,for%20clean%20hydrogen%20and%20derivatives>.

Many of the world's largest economies look at green hydrogen as both a decarbonization tool and an economic opportunity. Many countries, in anticipation of a global green hydrogen trade market, have published their own national hydrogen strategies. Most strategies look to utilize green hydrogen for domestic decarbonization, domestic manufacturing growth, industrial job creation, and as a source of revenue through exports.

These goals create an interesting dynamic where most countries look to become hydrogen exporters. Only Japan, Germany, and the United Kingdom discuss the need for green hydrogen imports to meet decarbonization goals. Although France, South Korea, and Mexico have not explicitly stated their import/export intentions, the remaining G20 nations have distinctly outlined the export of green hydrogen as a key objective. With expected hydrogen exporters outnumbering expected hydrogen importers, the global hydrogen market will likely be competitive.

This push towards domestic decarbonization and a robust export market is under

pinned by the ambitious production goals set by nearly all G20 countries, with most looking to multiply their current green hydrogen production capacity by over one hundred times by 2050. For instance, Argentina looks to produce over 5 million tons/year of green hydrogen by 2050, requiring a least 30 GW of electrolysis capacity. It currently has 0.8 megawatts (MW) of installed electrolysis capacity. This is not an isolated trend, with every G20 country (except for Mexico) looking to produce over 1 million tons/year of green hydrogen by 2050. Achieving these goals and staying in line with a 2050 net-zero emissions scenario will require significant increases in global electrolyzer production and additional renewable energy generation, on the order of a factor of eighty by 2030.

Table 2: The Current G20 Hydrogen Landscape

COUNTRY	H2 STRATEGY/ROADMAP	NUMBER OF PROJECTS	TOTAL INSTALLED ELECTROLYZER CAPACITY (MW)	EXPORT/IMPORT FOCUS IN H2 STRATEGY
ARGENTINA	Focus: Increase in domestic manufacturing, export markets, generation of jobs. Goal: Produce 5 million tons/year of green hydrogen by 2050, 80% of which will be exported.	Feasibility study (3), Concept (4), Operational (1)	0.828	Export
AUSTRALIA	Focus: Hydrogen as next large energy export, domestic decarbonization in transport, gas networks, etc. Goal: Upwards of 300 billion AUD in investment opportunity, major global player by 2030.	Feasibility study (65), Concept (47), FID/Construction (19), Operational (9), DEMO (6), Other/Unknown (1)	3.67	Export
BRAZIL	Focus: Creation of export opportunities, domestic decarbonization, manufacturing and job growth. Goal: Install low carbon hydrogen plants by 2025, most competitive producer/exporter by 2030, consolidate hydrogen hubs by 2035.	Feasibility study (7), Concept (12), Operational (2), DEMO (2), FID/Construction (1)	3.9	Export
CANADA	Focus: Encourage early deployment in mature applications. Incentivize production and demand via Clean Fuel Standard. Goal: Generate over 350,000 jobs, hydrogen to deliver 30% of Canada's total end-use energy by 2050.	Feasibility study (15), Concept (11), Operational (10), FID/Construction (10), DEMO (5)	23.7	Export

CHINA	Focus: Domestic decarbonization in transport, electricity generation/storage, etc. Goal: Produce 100,000 – 200,000 metric tons/year by 2025, 50,000 hydrogen vehicles by 2050.	Operational (21), Concept (15), Feasibility study (8), FID/Construction (31), DEMO (5), Decommissioned (1) Decommissioned (1)	581.29	N/A
FRANCE	Focus: Decarbonize industry, hydrogen mobility, research/innovation. Goal: Produce 700,000 tons/year and target price of \$1.6/kg by 2030.	Operational (17), Concept (35), Feasibility study (40), FID/Construction (19), DEMO (14)	19.9	N/A
GERMANY	Focus: Domestic decarbonization in heavy industry and transport sectors, building hydrogen transport networks. Goal: Decarbonize heavy industry and long-distance transport by 2030, 1,800 km of new/converted hydrogen pipelines by 2027-28.	Operational (54), Concept (31), Feasibility study (53), Demo (33), FID/Construction (25)	82.8	Import
INDIA	Focus: Become leading exporter of green hydrogen, domestic decarbonization, development of domestic manufacturing. Goal: Produce 5 million tons/year and 125 GW of additional renewable energy capacity by 2030.	Concept (22), Demo (3), FID/Construction (9),	13.36	Export
INDONESIA	Focus: Initial use for transportation sector by 2031, high temperature industrial heating processes by 2041, build export market Goal: Net zero emissions by 2060.	Concept (3), Feasibility study (3)	N/A	Export
ITALY	Focus: Decarbonize, heavy industry, long distance transport, refineries, and gas grid. Goal: 2% hydrogen penetration in final energy demand by 2030, 20% by 2050.	Operational (1), Concept (2), Feasibility study (2), DEMO (1)	3.95	Export
JAPAN	Focus: Decarbonize heating, transport, and production of e-fuel/e-methane. Explicit focus on safety. Goal: Consume 3 million tons per year by 2030, 20 million by 2050, and 12 million tons of ammonia per year by 2040.	Operational (6), Concept (11), Feasibility study (17), FID/Construction (5), DEMO (3), Other/Unknown (1)	18.91	Import
MEXICO	N/A	Concept (2), Feasibility study (2)	N/A	N/A

RUSSIA	Focus: Export to Europe and Asia. Goal: Export 200,000 tons by 2024, 2 – 12 million tons by 2035, and 15 – 50 million tons by 2050.	Operational (4), Concept (4), Feasibility study (7), FID/ Construction (2)	1	Export
SAUDI ARABIA	Focus: Global hydrogen supplier, diversify away from crude oil. Goal: Produce 2.9 million tons per year by 2030, 4 million tons per year before 2035.	Operational (4), Concept (4), Feasibility study (7), FID/ Construction (2)	N/A	Export
SOUTH AFRICA	Focus: Decarbonize economy, create economic opportunity domestically and through exports. Goal: Produce 500,000 tons per year by 2030, deploy 10 GW of electrolyzer capacity by 2030, and 15 GW by 2040.	Operational (2), Concept (6), Feasibility study (10), FID/ Construction (1)	3.5	Export
SOUTH KOREA	Focus: Decarbonizing transport through hydrogen cars and buses, utilize fuel cells for power generation. Goal: Produce 30,000 hydrogen-powered vehicles by 2030, build 70 liquid hydrogen refueling stations, and compose 7.1% of total energy mix with clean hydrogen.	Operational (6), Concept (9), Feasibility study (11), FID/ Construction (6), DEMO (2)	4	N/A
TURKEY	Focus: Domestic decarbonization in hard-to-abate sectors, blending into gas grid, export surplus. Goal: Reduce the levelized cost of hydrogen to \$2.4 by 2035 and \$1.2 by 2050, 5-20% hydrogen blend until 2025, then 20 – 100% between 2025 and 2040.	Operational (1), Concept (3), Feasibility study (2), FID/ Construction (1)	0.275	Export
UNITED KINGDOM	Focus: Domestic decarbonization, create jobs and economic growth. Goal: 10 GW of low carbon hydrogen production by 2030.	Operational (15), Concept (26), Feasibility study (36), FID/ Construction (12), DEMO (4), Other/ Unknown (1)	10.39	Import
UNITED STATES	Focus: Domestic decarbonization in “high-impact end uses,” reduce cost of green hydrogen, focus on regional networks. Goal: Produce 10 million tons/ year of clean hydrogen by 2030, establish green hydrogen costs of \$2/kg by 2026 and \$1/kg by 2031, deploy regional clean hydrogen hubs	Operational (27), Concept (38), Feasibility study (55), FID/ Construction (27), DEMO (9), Other/ Unknown (2)	52.95	Export

Source: Data sources for each country entry can be found in Appendix B. Note that this table does not include the European Union or African Union.

3. INDIA'S HYDROGEN GOALS

India recently launched its National Green Hydrogen Mission, aiming to become a leader in the global green hydrogen market. The Green Hydrogen Mission primarily positions India as a global green hydrogen exporter, designating 70 percent of its 2030 green hydrogen production targets (around 3.5 million tons) for export and the remaining hydrogen for domestic decarbonization. The mission includes reductions in taxes and tariffs along with financial incentives totaling nearly \$2 billion to increase domestic electrolyzer manufacturing and green hydrogen production.¹²

Table 3: Assumptions and Parameters to Estimate Cost

PARAMETER	VALUE
Cost of Capital (Discount Rate) ¹³	5.9% -> 15%
Electrolyzer Efficiency (kWh/kg) ¹⁴	50
Solar Costs (for India, \$/kW) ¹⁵	Capital/Equipment Costs: 590 Operating Costs: 9.6
Electrolyzer Costs (Low, Medium, High Scenarios 2022 \$/kW) ¹⁶	Capital/Equipment Costs: 1150, 1450, 1800 Operating Costs: 17, 21, 26
Lifetime (years) ¹⁷	20

Source: Methodology and full input parameters can be found in Appendix A and Appendix C.

This study computes green hydrogen production costs at locations across India. Cost estimates use inputs unique to each location at a geographical resolution of 0.25 degrees latitude by 0.25 degrees longitude (approximately 550 km²). At each grid cell, these inputs include parameters such as renewable energy generation from wind and solar patterns based on hourly values averaged over 15 years, capital and operating costs, cost of capital, and plant lifetime. Then the assessment finds the lowest cost of hydrogen and the optimal ratio between renewable energy generation and electrolyzer capacity to generate green hydrogen.

For India, only solar generation due to greater accessibility throughout the region is considered.¹⁸ Grid-connected electrolyzers are not considered to be sources of green hydrogen, as in order to ensure that embodied grid emissions (grid electricity gener-

¹² "Nearly 50 MMT per annum of CO₂ emissions can be averted through production and use of Green Hydrogen as targeted under National Green Hydrogen Mission: New & Renewable Energy Minister," PIB Delhi, August 1, 2023, <https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1944758>.

¹³ IRENA, "Data Appendix: Country and technology-specific real after-tax WACC assumptions for 2021," accessed on November 17, 2023, https://mc-cd8320d4-36a1-40ac-83cc-3389-cdn-endpoint.azureedge.net/-/media/Files/IRENA/Agency/Publication/2023/May/IRENA_Cost_of_financing_renewable_power_Appendix_2023.pdf?rev=0c9288a22ab44676a1fd5b8ca0c774e8.

¹⁴ Gunther Glenk, Philip Holler, and Stefan Reichelstein, "Advances in Power-to-Gas Technologies: Cost and Conversion Efficiency," MIT CEEPR, April 2023, <https://ceep.mit.edu/wp-content/uploads/2023/04/MIT-CEEPR-WP-2023-09.pdf>.

¹⁵ Michael Taylor et al., "Renewable power generation costs in 2021," IRENA, July 2022, <https://www.irena.org/publications/2022/Jul/Renewable-Power-Generation-Costs-in-2021>.

¹⁶ Herib Blanco et al., "The Future of Hydrogen," IEA, June 2019, <https://www.iea.org/reports/the-future-of-hydrogen>.

¹⁷ Muhammad Arfan et al., "Life Cycle Assessment and life cycle costing of hydrogen production from biowaste and biomass in Sweden," Energy Conversion Management, September 2023, <https://doi.org/10.1016/j.enconman.2023.117262>.

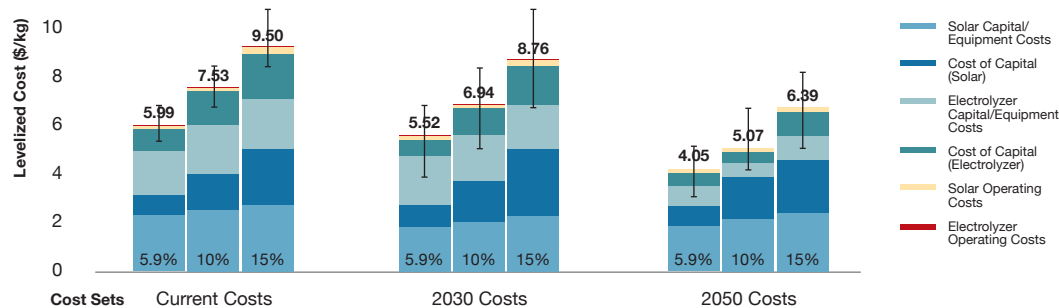
¹⁸ The World Bank, "Global Solar Atlas," accessed on December 5, 2023, <https://globalsolaratlas.info/>.

ated from coal or gas) are accounted for, strict regulations must be put in place, like the regulations instituted by the United States in from the Inflation Reduction Act.^{6 19} Depending on an electricity grid's generation profile, a direct grid-electrolyzer system could result in more emissions than grey and blue hydrogen generation.²⁰

Furthermore, the author chose to analyze the median instead of the minimum cost of production to represent overall production costs more accurately throughout the region. Importantly, this analysis does not consider the effect of any domestic subsidies or tax incentives currently in place. Costs presented are purely production and not landed costs. It should be noted that these costs are likely underestimates of true production costs as parameters such as the cost of water, land permitting, and construction time are not considered. Also note that all costs presented are levelized cost, computed by taking the total capital and operating costs divided by the total amount of hydrogen produced, both discounted over the lifetime of the hydrogen generation plant. Additional costs accrued through compression, storage, and transportation would likely increase the levelized cost.²¹

4. CAPITAL AND ITS COST ARE THE LARGEST DRIVERS OF SOLAR-POWERED GREEN HYDROGEN PRODUCTION COSTS IN INDIA

Figure 2: Median Green Hydrogen Production Cost Sets



This is a breakdown of median green hydrogen production costs throughout India at various discount rates. Each bar shows the median production cost at different costs of capital (5.9 percent, 10 percent, 15 percent). Error bars represent levelized cost given higher and lower electrolyzer cost estimates. Solar and electrolyzer capital expenditures make up most costs. As electrolyzer costs fall, its proportion of total levelized cost falls accordingly.

Source: Author's analysis.

Production costs are estimated under three different electrolyzer cost scenarios: neutral/business-as-usual, an optimistic scenario with lower electrolyzer cost assumptions, and a pessimistic scenario with higher electrolyzer cost assumptions across various costs of capital (Figure 2). Costs are projected in three different years: 2022, 2030 and 2050. In 2022, under the lowest assumptions for cost of capital, estimated median green hydrogen costs in India are approximately \$6 per kilogram (\$5.30–6.70 per kg). By 2030, estimated costs fall to \$5.50 (\$4.30–6.70 per kg). By 2050, costs fall to \$4.05 (\$3.30–4.80 per kg). These cost

¹⁹ "Section 45V Credit for Production of Clean Hydrogen; Section 48(a)(15) Election To Treat Clean Hydrogen Production Facilities as Energy Property," IRS, December 26, 2023, <https://www.federalregister.gov/documents/2023/12/26/2023-28359/section-45v-credit-for-production-of-clean-hydrogen-section-48a15-election-to-treat-clean-hydrogen>.

²⁰ Wilson Ricks, Qingyu Xu, and Jesse D Jenkins, "Minimizing emissions from grid-based hydrogen production in the United States," IOP Science, January 2023, <https://iopscience.iop.org/article/10.1088/1748-9326/acab55>.

²¹ "Hydrogen Market Report," FIPI, October 23, 2023, <https://www.fipi.org.in/assets/pdf/downstream-report/Hydrogen%20Market%20Report.pdf>.

ranges align with estimates seen in a number of previous analyses.²² The main assumptions to derive these figures are outlined in Table 3 with a full description in Appendix A.

The largest drivers of production cost are cost of capital and capital equipment cost of solar energy systems and electrolyzers. In 2022, the analysis suggests that solar and electrolyzer capital costs account for approximately two-thirds of overall levelized cost, evenly split between equipment costs and the cost of capital (discount rate). Operating costs of solar systems and electrolyzers for green hydrogen production are less than 5 percent of total levelized costs.

Even with optimistic reductions in capital equipment costs, the cost of capital required to finance that equipment must also decrease. Alone, falling capital costs result in cost estimates of \$4 per kg by 2050, two times more expensive than 2022 conventionally produced grey hydrogen costs (Table 1). Higher costs of capital (10 percent and 15 percent) increase this disparity, resulting in production cost estimates 20-60 percent higher than the lowest cost of capital (5.9 percent), creating a cost premium between 2 – 3.5 times current grey hydrogen prices.

Cost of capital is consequently a limiting factor in green hydrogen production costs. Cost of capital for green hydrogen projects can vary widely, often because of diverse risk factors.²³ These include:

Technology Risk: Green hydrogen production is a new and growing industry. Electrolyzer costs are uncertain, safety standards and definitions range from country to country, or are non-existent. The lack of large-scale projects means that uncertainty also exists regarding maintenance time, electrolyzer efficiencies and lifetimes, repair costs, and other unforeseen challenges. These challenges increase the technology risk associated with project financing, resulting in significantly higher costs of capital and subsequent production costs.

Demand Risk: Securing both long and short-term off-takers for hydrogen production is one of the largest barriers for projects looking for funding.²⁴ Without an established selling price, hydrogen production is subject to market volatility and the uncertainties of fluctuating demand trends.

Policy Risk: Policy incentives, governmental regulations, and clean hydrogen standards are evolving. The absence or presence of these incentives and regulations can swing the price difference between grey and green hydrogen significantly, resulting in added uncertainty for investors or project developers.

The presence of these risk components can result in costs of capital upwards of 20 percent higher than the assumed cost of capital here.²⁵

²² Tirtha Biswas, Deepak Yadav, and Ashish Guhan Baskar, "A Green Hydrogen Economy for India," CEEW, December 2020, <https://www.ceew.in/sites/default/files/CEEW-A-Green-Hydrogen-Economy-for-India-14Dec20.pdf>.

²³ Moongyung Lee and Deger Saygin, "Financing cost impacts on cost competitiveness of green hydrogen in emerging and developing economies," OECD, November 2023, [https://one.oecd.org/document/ENV/WKP\(2023\)](https://one.oecd.org/document/ENV/WKP(2023)).

²⁴ Lee, "Financing cost impacts on cost competitiveness of green hydrogen in emerging and developing economies."

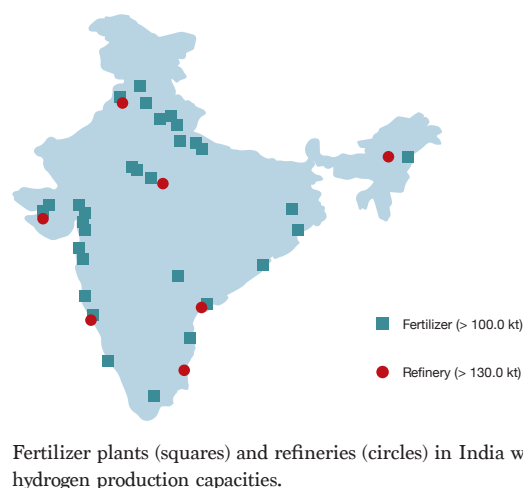
²⁵ Lee, "Financing cost impacts on cost competitiveness of green hydrogen in emerging and developing economies."

5. DECARBONIZING REFINERIES AND FERTILIZERS BY TRANSITIONING TO GREEN HYDROGEN

India could partly address demand risk by focusing its green hydrogen efforts on two sectors with tangible, established hydrogen demand: fertilizers and petroleum refining, where the country ranks globally, third and fourth, respectively.^{26 27 28 29}

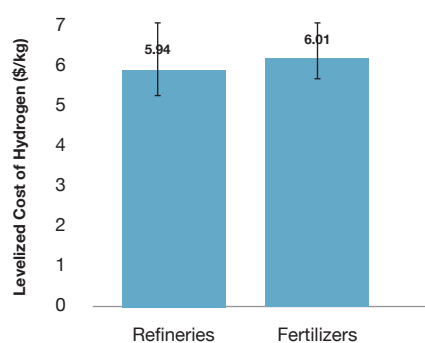
These industries, which currently use approximately 6 million tons of grey hydrogen, represent an implicit demand baseline, making it easier for producers to secure off-take agreements and hedge project risk.³⁰ However, this transition from grey to green hydrogen will incur a steep cost premium.

Figure 3: Fertilizer plants and Refineries in India



Source: Author's analysis.

Figure 4: Levelized Cost Comparison



Median levelized hydrogen costs of fertilizer and refinery plants across India. Error bars show minimum and maximum values. The cost of capital is set to 5.9 percent.

Source: Author's analysis.

Currently, fertilizer plants and refineries sit throughout India with estimated green hydrogen production costs ranging between \$5.50 and \$7.30 per kg (see Figures 5 and 6).^{31 32} Compared to grey hydrogen, the cost of green hydrogen production using solar energy at locations of fertilizer plants and refineries is three- to four-fold higher. These costs do not consider the expenses involved in retrofitting existing plant designs or generating new workflows to integrate green hydrogen generation systems. Moreover, production at coastal plants can likely achieve reduced costs by utilizing wind in conjunction with solar generation.

²⁶ "Production volume of nitrogen fertilizer worldwide in 2018, by country," Statista, April 2021, <https://www.statista.com/statistics/1252656/nitrogen-fertilizer-production-by-country>.

²⁷ "Crude oil refinery capacity worldwide in 2010 and 2022, by major country," Statista, June 2023, <https://www.statista.com/statistics/273579/countries-with-the-largest-oil-refinery-capacity/>.

²⁸ Friedrich et al., "This Interactive Chart Shows Changes in the World's Top 10 Emitters," WRI, March 2023, <https://www.wri.org/insights/interactive-chart-shows-changes-worlds-top-10-emitters>.

²⁹ "Renewable Energy Sector," InvestIndia, February 23, 2024, <https://www.investindia.gov.in/sector/renewable-energy>.

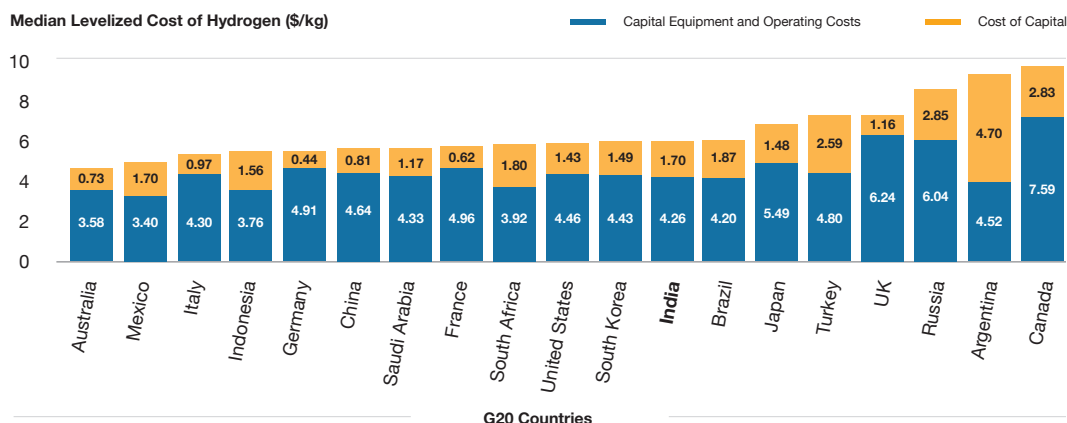
³⁰ Gül et al., "Global Hydrogen Review 2023."

³¹ "Annual Report 2020-2021," Department of Fertilizers, 2021, <https://www.fert.nic.in/sites/default/files/2020-08/2022-08/Annual%20report%20of%20Department%20of%20Fertilizer%202020-2021.pdf>.

³² Hall et al., "The Potential Role of Hydrogen in India," 2020, https://www.teriin.org/sites/default/files/2021-07/Report_on_The_Potential_Role_of_%20Hydrogen_in_India.pdf.

6. ABSENT INCENTIVES INDIA'S GREEN HYDROGEN PRODUCTION COSTS ARE HIGHER THAN MANY G20 COUNTRIES

Figure 5: Comparison of G20 Median Levelized Costs of Production Using Solar Renewable Energy Resources



India ranks as the 12th cheapest median production cost. Total costs are split into two components, cost of capital and capital equipment/operating cost. Cost of capital in countries with a lack of solar resources has increased price impact as countries must overbuild solar systems to compensate for the lack of resources.

Source: Author analysis. See Appendixes A, B, and C.

Comparing India's green hydrogen production costs with the rest of the G20 underscores the importance of the cost of capital. Estimated costs in Figure 7 include two components: capital and operating costs and the cost of capital. While capital and operating costs to produce green hydrogen may be lower in some countries like India, many of these countries incur a higher cost of capital, increasing their overall cost of green hydrogen production.

Currently, India stands 12th in the G20 in terms of median green hydrogen production cost, at approximately \$6 per kg. This cost is 40 percent higher than the G20's least expensive producer, Australia. Crucially, India's production costs are higher than several countries positioning themselves hydrogen importers like Germany, due to a low cost of capital, and countries positing themselves exporters like Saudi Arabia, South Africa, and Australia. Factors such as access to lower discount rates (notable in Germany and France) and better median solar resources (seen in Australia, Saudi Arabia, and China) drive these cost differences. It is important to note that this analysis is based solely on solar energy. Countries like Argentina, Canada, Russia, and France, which have greater access to wind resources, could significantly reduce production costs by utilizing wind energy.

These cost differences pose challenges for India's hydrogen exporter goal. Several importers and exporters within the G20 have lower production costs, meaning that competing in an international hydrogen market will require significant efforts to achieve cost competitiveness. To be a competitive exporter within the G20, India must reduce median production costs by about 30 percent. To be able to produce hydrogen at a cost cheaper than G20 importers (Germany), median production costs must be reduced by around 10 percent. While India has several tax subsidies and financial incentives in place through the Green Hydrogen Mission, several other G20 countries have done the same. The United States has released its 45V hydrogen tax credit, offering up to \$3.00 per kilogram of green hydrogen

produced, Argentina is providing a complete tax break for green hydrogen produced within the next decade, and Australia is providing over \$1 billion dollars in subsidies.^{33 34} For India to be cost competitive, its financial incentives and policy mechanisms must not only match these international efforts but exceed them in order to bridge current cost disparity.

6. CONCLUSION AND RECOMMENDATIONS

India's role within the growing global green hydrogen value chain is uncertain. Estimated production costs of off-grid solar-based green hydrogen in India are higher than conventional hydrogen production methods and higher than in other countries. This poses challenges for domestic substitution and international export. Falling electrolyzer costs are not enough to make up this difference. Even in the case where blue and grey hydrogen prices stay constant, assumed capital and operation costs reductions by 2050 still result in green hydrogen being 20 percent and 70 percent more expensive than its blue and grey counterparts, respectively. Under current cost conditions, replacing a kilogram of grey hydrogen with a kilogram of green hydrogen would require a cost reduction of \$3.30 per kilogram in India. If India were to directly subsidize this cost difference and meet its goal of 5 million tons of green hydrogen by 2030, it would require \$16.5 billion per year, significantly more than the current \$2.4 billion outlay. Importantly, a majority of these incentives are not slated for direct subsidies and are instead meant to encourage research and development primarily to reduce electrolyzer capital costs. However, our analysis shows that even with all-in electrolyzer costs of \$200/kilowatt (kW), levelized costs are approximately 70 percent higher than current grey hydrogen prices. Thus, to achieve cost parity, these incentives must also reduce other cost drivers, such as the cost of capital, or make grey hydrogen more expensive through a price on carbon emissions.

India may be able to reduce its cost of capital by refocusing its policy and financial mechanisms towards incentivizing domestic decarbonization efforts for two major reasons: guaranteed off-takers and reduced uncertainty. By looking to replace grey hydrogen production in refineries and fertilizer plants (many of which are owned by state-owned enterprises), India can build out and de-risk projects while not having to secure outside demand. With only 10 percent of the clean hydrogen capacity planned by 2030 having an off-taker, India's domestic demand acts as a clear advantage.³⁵ Thus, to capitalize on this while simultaneously reaching its decarbonization goals, India could look to focus capital away from a high-risk, competitive, and uncertain export market and instead shift towards decarbonizing its refineries and fertilizer plants. Such a shift could shed uncertainty shrouding hydrogen production, possibly resulting in lower costs of capital which can have significant effects on overall levelized production costs.

The effectiveness that guaranteed demand and off-take offers has been seen before,

³³ "Section 45V Credit for Production of Clean Hydrogen; Section 48(a)(15) Election To Treat Clean Hydrogen Production Facilities as Energy Property," IRS.

³⁴ Charles Newbery, "Argentina to submit bill providing incentives to produce hydrogen for export," S&P Global, May 19, 2023, <https://www.spglobal.com/commodityinsights/en/market-insights/latest-news/energy-transition/051923-argentina-to-submit-bill-providing-incentives-to-produce-hydrogen-for-export#:~:text=The%20bill%20calls%20for%20providing,a%20draft%20of%20the%20bill.>

³⁵ "Hydrogen offtake is tiny but growing," BNEF, November 14, 2023, <https://about.bnef.com/blog/hydrogen-offtake-is-tiny-but-growing/#:~:text=Many%20companies%20have%20plans%20to,for%20clean%20hydrogen%20and%20derivatives.>

specifically with South Africa's Renewable Energy Independent Power Producer Procurement Program. By introducing a tender process for renewable energy companies that, when selected, ensured a 20-year purchase price allocation (PPA) with Eskom, South Africa's state utility, the country was able to incentivize renewable energy generation at reduced risk.³⁶ Since 2011, the program has created 64 new renewable energy independent power producers (IPPs) and \$14 billion in investment to fund the creation of 3,922 MW of various renewable energy capacity.³⁷ India is taking an analogous approach to de-risking hydrogen investment and development by investigating to see the potential for a green hydrogen mandate in fertilizer and refineries.³⁸

Access to lower costs of capital will also accelerate India's pursuit to become a leading global hydrogen exporter. Cost comparisons show that Indian green hydrogen while potentially cost competitive against other G20 countries due to abundant solar resources, pays a penalty for higher cost of capital. This makes economic viability within the global export market without financial incentives difficult. If green hydrogen projects in India pay lower costs of capital, they can begin to make up the cost difference between India and the rest of the G20. While a focus on domestic production and decarbonization would likely decrease risk, other financial incentives such as concessional capital could have similar effects.

Appendix A: Plant and Financial Input Parameters

This model accounts for electrolyzer degradation and stack replacement costs across the lifetime of the hydrogen plant. The number of stack replacements and the overall cost associated with each replacement is aggregated into the overall electrolyzer CAPEX. The source of data for each parameter can be found in Appendix C. This analysis contains limitations and uncertainty. All cost estimates described are purely production costs and do not include the cost of storage or transport. Land accessibility is not addressed and would likely increase estimated production costs. Renewable energy generation and electrolyzers are assumed to be co-located, thus transmission costs are not considered. Due to the lack of reliable electrolyzer operational costs at different years and in different scenarios, operational expenditure costs in future years and scenarios are scaled from current (2022) costs by the percentage increase or decrease in capital costs. Significant uncertainty exists within true electrolyzer cost and project discount rate estimates due to the lack of developed, large-scale green hydrogen projects. Furthermore, the inclusion of wind or a combination of solar and battery storage as renewable energy resources would provide increased capacity factors and thus, lower production costs. Finally, true production costs estimates will require an analysis of compression, storage, and transport costs. Further work of this analysis may include wind solar hybrid locations.

PARAMETER	VALUE	SOURCE
India Cost of Capital (Discount Rate)	5.9% -> 15%	IRENA (2022)
Electrolyzer Efficiency (kWh/kg)	50	CEEPR (2023)
Solar Costs (for India, \$/kW)	Capital/Equipment Costs: 590 Operating Costs: 9.6	IRENA (2021)
Electrolyzer Capital Costs (Low, Medium, High Scenarios 2022 \$/kW)	1100, 1450, 1800	IEA (2019)
Electrolyzer Operating Costs (Low, Medium, High Scenarios 2022 \$/kW)	17, 22, 28	IEA (2023)
Electrolyzer Operating/Capital Costs (Low, Medium, High Scenarios 2030 \$/kW)	Capital: 650, 1225, 1800 Operating: 10, 18, 28	IEA (2019)
Electrolyzer Operating/Capital Costs (Low, Medium, High Scenarios 2050 \$/kW)	Capital: 200, 550, 900 Operating: 3, 9, 14	IEA (2019)
Lifetime (years)	20	Arfan et al. (2023)
G20 Cost of Capital (Discount Rate)	Varies	IRENA (2022)
Global Solar Costs	Varies	IRENA (2021)
Renewable Energy Data (Solar Irradiance)	Varies	ERA 5 Reanalysis (2007 – 2022)

Electrolyzer Stack Degradation Rate (% per year)	2	Energy Proceedings (2022)
Allowed Electrolyzer Degradation Before Replacement (%)	70	
Electrolyzer Replacement Cost (% of Initial CAPEX)	30	

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INDIA: THE RISING POWER IN GLOBAL SOLAR PHOTOVOLTAIC SUPPLY CHAINS

BY GREGORY WISCHER

SUMMARY

Electricity from solar photovoltaic modules is an important and increasing part of many countries' energy mix. Currently, China and Chinese companies dominate the manufacturing supply chain for solar photovoltaic technology, from the polysilicon to the solar modules. Yet, countries are increasingly concerned about overdependence on China and thus are seeking to diversify the sourcing of their solar photovoltaic inputs. India has existing production and latent potential to serve as an alternative supplier to China in the solar photovoltaic supply chain, especially for solar cells and modules. International cooperation, like concessional financing, could help Indian companies develop upstream production capacity in polysilicon, ingots, and wafers, and international cooperation could also help finance renewable energy projects, which could power solar-related manufacturing projects and reduce their electricity costs.

1. INTRODUCTION

Electricity from solar photovoltaic (PV) modules is a key, and growing, part of many countries' energy mix. Converting sunlight into electricity benefits many countries by growing electricity access, supporting decarbonization efforts, and increasingly, lowering electricity costs. Geopolitically too, countries that deploy solar PV modules strengthen their energy security if it displaces imports of other energy sources like coal or natural gas. Thus, many countries, including India, Brazil, China, the European Union, and the United States, are pursuing wide-scale deployment of solar PV modules.¹ In fact, solar PV technology represented 56 percent of all global electricity capacity additions in 2022, and solar PV's installed power capacity is projected to be the largest of any power source by 2027, surpassing coal.²

Currently, the manufacturing supply chain for solar PV technology is dominated by China and Chinese companies, many of which also operate in Southeast Asia. In 2023, China's share of global manufacturing capacity in each solar PV supply chain segment was 75 percent or more, and its future share of global manufacturing capacity across the solar PV supply chain is forecasted to range between 80 percent to 95 percent.³ Most countries' solar PV policies prioritize wide-scale solar deployment and thus low-cost solar modules, making them heavily reliant on Chinese-produced inputs. China's dominance can be partly attributed to significant state support, which has given Chi-

¹ International Energy Agency, "Renewables 2023: Analysis and Forecast to 2028," January 2024, 7–9, https://iea.blob.core.windows.net/assets/96d66a8b-d502-476b-ba94-54ffda84cf72/Renewables_2023.pdf; and Shayak Sengupta, "Diversifying Solar Supply Chains: Costs and the Case for International Coordination," ORF America, July 2023, <https://orfamerica.org/newresearch/solar-supply-chains-diversity-coordination>.

² Harrison Dreves, "At a Glance: How Renewable Energy Is Transforming the Global Electricity Supply," National Renewable Energy Laboratory, September 26, 2023, <https://www.nrel.gov/news/program/2023/how-renewable-energy-is-transforming-the-global-electricity-supply.html>; and International Energy Agency, "Renewables 2022: Analysis and Forecast to 2027," December 2022, 11, <https://iea.blob.core.windows.net/assets/ada7af90-e280-46c4-a577-df2e4fb44254/Renewables2022.pdf>. Solar PV is forecasted to represent the world's largest capacity additions for renewable electricity from 2023 to 2028. See IEA, "Renewables 2023," 14.

³ "Solar PV Manufacturing Capacity by Country and Region, 2021," International Energy Agency, updated July 5, 2022, <https://www.iea.org/data-and-statistics/charts/solar-pv-manufacturing-capacity-by-country-and-region-2021>; IEA, "Renewables 2023," 9.

nese companies lower production costs versus foreign competitors.⁴

However, relying on a few suppliers from one country poses risks, including supply chain risks. If Chinese solar companies face production issues or if the Chinese government restricts exports of solar PV inputs, many countries would be unable to deploy cheap solar at scale, undermining targets for energy access and renewable energy deployment. Importantly, high reliance on China for solar PV inputs also subverts labor and environmental protections. China's Xinjiang Uyghur Autonomous Region produces nearly 45 percent of the world's polysilicon for solar PV modules, and all of the region's major polysilicon manufacturers have participated in Chinese government programs associated with forced labor.⁵

Thus, several countries, especially those concerned with overdependence on China like the United States, are seeking to diversify their sourcing for inputs in the solar PV supply chain. India is one such alternative supplier, and it too is seeking to diversify its solar PV supply chain.⁶ Already, Indian solar module exports have increased by more than five times from fiscal year 2022 to 2023, and the overwhelming majority of India's solar module exports go to the United States.⁷ Moreover, India's solar-related production-linked incentive (PLI) scheme, which provides incentives to companies to produce solar-related components in India, will further bolster India's production and export capacity across the solar PV supply chain.⁸ Consequently, India could become an alternative global supplier to China for solar PV cells and modules, and potentially upstream inputs like polysilicon, ingots, and wafers too.

⁴ International Energy Agency, "Special Report on Solar PV Global Supply Chains," July 2022, 7, 107, <https://iea.blob.core.windows.net/assets/d2ee601d-6b1a-4cd2-a0e8-db02dc64332c/SpecialReportonSolarPVGlobalSupplyChains.pdf>; and "China's Solar Production Costs Fall by 42% in Last Year," Wood Mackenzie, December 14, 2023, <https://www.woodmac.com/press-releases/chinas-solar-production-costs-fall-by-42-in-last-year/>. For a comprehensive analysis of the Chinese government's policies on solar PV manufacturing, see Yueqi Zhang et al., "Evolution of Solar Photovoltaic Policies and Industry in China," *IOP Conference Series: Earth and Environmental Science* 651 (2021): 1–22, <https://doi.org/10.1088/1755-1315/651/2/022050>.

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⁶ IEA, "Special Report on Solar PV Global Supply Chains," 58–62; Sengupta, "Diversifying Solar Supply Chains."

⁷ Jyoti Gulia et al., "India's Photovoltaic Manufacturing Capacity Set to Surge," Institute for Energy Economics and Financial Analysis, JMK Research & Analytics, April 2023, 6, <https://ieefa.org/sites/default/files/2023-04/India%27s%20Photovoltaic%20Manufacturing%20Capacity%20Set%20to%20Surge-April%202023%20%283%29.pdf>; David Feldman et al., "Spring 2023: Solar Industry Update," National Renewable Energy Laboratory, April 27, 2023, 12, <https://www.nrel.gov/docs/fy23osti/86215.pdf>.

⁸ "Production Linked Incentive (PLI) Scheme: National Programme on High Efficiency Solar PV Modules," Ministry of New and Renewable Energy, Government of India, accessed April 4, 2024, <https://mnre.gov.in/production-linked-incentive-pli/>. For the successful bidders under the PLI schemes, see "List of Successful Bidders under Tranche-I of PLI Scheme 'National Programme on High Efficiency Solar PV Modules'," Ministry of New and Renewable Energy, Government of India, accessed April 4, 2024, <https://cdnbbsr.s3waas.gov.in/s3716e1b8c6cd17b771da77391355749f3/uploads/2023/08/2023080887.pdf>; and "Capacity Allocation under RfS for Selection of Solar PV Module Manufacturers for Setting up Manufacturing Capacities for High Efficiency Solar PV Modules in India under the Production Linked Incentive Scheme (Tranche-II)," Ministry of New and Renewable Energy, Government of India, accessed April 4, 2024, <https://cdnbbsr.s3waas.gov.in/s3716e1b8c6cd17b771da77391355749f3/uploads/2023/08/2023080844.pdf>. The Institute for Energy Economics and Financial Analysis and JMK Research & Analytics estimate that by 2026, India's production capacity for polysilicon will be 36 gigawatts (GW), ingots and wafers will be 56 GW, cells will be 59 GW, and modules will be 110 GW. See Gulia et al., "India's Photovoltaic Manufacturing Capacity," 6.

2. THE SOLAR PV SUPPLY CHAIN: CONTEXTUALIZING INDIA

The most common type of solar PV module is the crystalline silicon module.⁹ The other major type is the cadmium telluride thin-film PV module, but it comprises less than 5 percent of global solar PV production.¹⁰ For crystalline silicon modules, the core material is indeed silicon.¹¹ The supply chain begins with converting silica minerals like quartzite into metallurgical-grade silicon, which is then transformed into polysilicon. This polysilicon is next turned into an ingot that is cut and sliced into wafers. To create a solar cell, the wafers are cleaned, textured, and doped with various gases, liquids, and materials. Solar cells are then assembled to form a solar PV module. This supply chain can be represented in the following four-step process:

1. Polysilicon
2. Ingots/wafers
3. Cells
4. Modules

Figure 1: The Crystalline Silicon PV Module Supply Chain



Source: Paul Basore and David Feldman, “Solar Photovoltaics: Supply Chain Deep Dive Assessment,” Solar Energy Technologies Office, U.S. Department of Energy, February 24, 2022, iii, <https://www.energy.gov/sites/default/files/2022-02/Solar%20Energy%20Supply%20Chain%20Report%20-%20Final.pdf>.

While China has commanding production shares in all segments of the solar PV supply chain as shown in Figure 2, India is seeking to increase its production capacity.¹² In 2023, China produced approximately 91 percent of the world’s polysilicon for solar PV modules, while India had 0 percent of global production capacity for polysilicon in 2024, which is partly attributable to India’s high energy costs.¹³ The most common process for producing polysilicon is the Siemens process, and it uses extreme heat and consumes significant amounts of electricity, necessitating low-cost electricity.¹⁴ In fact,

⁹ IEA, “Special Report on Solar PV Global Supply Chains,” 13.

¹⁰ IEA, “Special Report on Solar PV Global Supply Chains,” 13.

¹¹ Crystalline silicon modules contain other materials, like silver and aluminum, but the core material is silicon. The production process also uses materials that do not constitute the final crystalline silicon modules, such as the high-purity quartz crucibles used to produce monocrystalline silicon ingots. See IEA, “Special Report on Solar PV Global Supply Chains,” 19–21.

¹² IEA, “Special Report on Solar PV Global Supply Chains,” 7.

¹³ Phred Dvorak and Andrew Mollica, “Can the US Break China’s Grip on Solar?” *Wall Street Journal*, February 12, 2024, <https://www.wsj.com/business/can-the-u-s-break-chinas-grip-on-solar-18837af6>; “Solar PV Manufacturing Capacity by Component in India, 2021–2024,” IEA; Gulia et al., “India’s Photovoltaic Manufacturing Capacity,” 10; Monika Mondal, “India Wants Its Own Solar Industry but Has to Break Reliance on China First,” *Climate Home News*, January 30, 2024, <https://cleanenergyfrontier.climatechangenews.com/india-wants-own-solar-industry-break-reliance-china/>.

¹⁴ IEA, “Special Report on Solar PV Global Supply Chains,” 37; Murphy and Elimä, “In Broad Daylight,” 28.

electricity represents an estimated 40 percent of the production costs for polysilicon.¹⁵

China's production dominance continues into the production of ingots and wafers.¹⁶ As of 2024, China produced more than 97 percent of the world's ingots for solar PV modules.¹⁷ Presently, India does not have commercial production of silicon ingots, although Adani Solar announced in 2022 that it had produced a monocrystalline silicon ingot.¹⁸ The company aims to produce 10 gigawatts (GW) of ingot capacity by 2025, and by 2026, India may have an estimated ingot production capacity of 56 GW.¹⁹ China also produced 97 percent of the world's wafers for solar PV modules as of 2024, and again, India lacks commercial production of wafers for solar modules.²⁰ For both ingot and wafer manufacturing, raw materials like polysilicon represent 32 percent of costs, while energy comprises 28 percent of costs and labor constitutes 8 percent of costs.²¹ India's relatively high electricity costs contribute to its lack of ingot and wafer manufacturing too.²²

China controls most of the world's production of solar cells, an estimated 80 percent as of 2024.²³ In March 2023, India had 6.6 GW of production capacity for solar cells, representing less than 1 percent of global production capacity for solar cells.²⁴ Current Indian companies producing solar cells include, among others, Adani Solar, Jupiter, Premier Energies, Tata Power Solar, and Webel Solar.²⁵ India faces workforce challenges with solar cell manufacturing given a lack of skilled labor capable of installing and operating advanced equipment at solar cell plants.²⁶

Lastly, China holds a commanding share of global solar module production. In 2023,

¹⁵ IEA, "Special Report on Solar PV Global Supply Chains," 11.

¹⁶ Most production of ingots and wafers occurs at the same facilities.

¹⁷ Dvorak and Mollica, "Can the US Break China's Grip on Solar?"

¹⁸ Gulia et al., "India's Photovoltaic Manufacturing Capacity," 10; "Solar PV Manufacturing Capacity by Component in India, 2021–2024," IEA; "Adani Solar's Green Breakthrough Introduces India's First Large Sized Monocrystalline Silicon Ingot," Adani Solar, accessed April 4, 2024, <https://www.adanisolar.com/Newsroom/Media-Releases/Adani-Solars-green-breakthrough-Introduces-Indias-first-large-sized-monocrystalline-silicon-ingot>.

¹⁹ "Adani Solar's Green Breakthrough," Adani Solar. Indosol Solar, a special-purpose vehicle of Shirdi Sai Electricals, announced it would build 500 megawatts (MW) of ingot and wafer capacity by 2025, and an additional 9.5 GW by 2026. See "Indosol Solar Aims to Complete Rs 15,000 Crore Ingot to Module Unit by 2025," *Economic Times*, April 4, 2024, <https://energy.economictimes.indiatimes.com/news/renewable/indosol-solar-aims-to-complete-rs-15000-crore-ingot-to-module-unit-by-2025/109034630>; Uma Gupta, "Shirdi Sai Electricals Targets 10 GW of Ingot-to-Solar Module Capacity," *PV Magazine*, April 5, 2024, <https://www.pv-magazine.com/2024/04/05/shirdi-sai-electricals-targets-10-gw-of-ingot-to-solar-module-capacity/>; Gulia et al., "India's Photovoltaic Manufacturing Capacity," 6.

²⁰ Dvorak and Mollica, "Can the US Break China's Grip on Solar?"; Gulia et al., "India's Photovoltaic Manufacturing Capacity," 10.

²¹ Bhupesh Verma and Parveen Kumar, "Feasibility Analysis for c-Si PV Manufacturing in India," Center for Study of Science, Technology & Policy, 2018, 5, https://cstep.in/drupal/sites/default/files/2020-08/WP_SiPV%20Manufacturing_0.pdf.

²² Narendra Shiradkar, "Recent Developments in Solar Manufacturing in India," *Solar Compass* 1 (2022): 7–10, <https://doi.org/10.1016/j.solcom.2022.100009>; IEA, "Special Report on Solar PV Global Supply Chains," 11, 37, 119. Country-to-country differences in wafer manufacturing costs are largely attributable to differences in labor and electricity costs. See Michael Woodhouse, "Crystalline Silicon Photovoltaic Module Manufacturing Costs and Sustainable Pricing: 1H 2018 Benchmark and Cost Reduction Road Map," National Renewable Energy Laboratory, February 2020, 20, <https://www.nrel.gov/docs/fy19osti/72134.pdf>.

²³ Dvorak and Mollica, "Can the US Break China's Grip on Solar?"

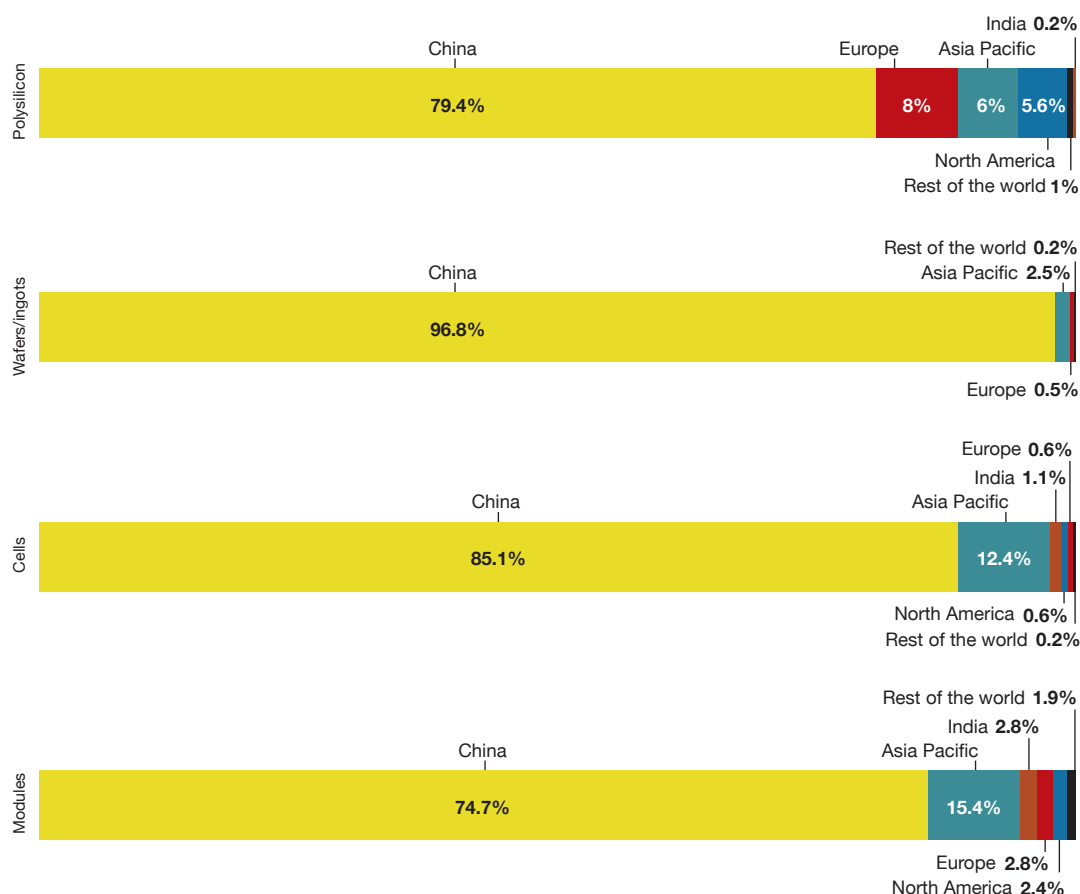
²⁴ Gulia et al., "India's Photovoltaic Manufacturing Capacity," 10; "Solar PV Manufacturing Capacity by Component in India, 2021–2024," IEA.

²⁵ Gulia et al., "India's Photovoltaic Manufacturing Capacity," 15.

²⁶ Gulia et al., "India's Photovoltaic Manufacturing Capacity," 7.

China produced 83 percent of the world's solar PV modules.²⁷ Countries in Southeast Asia, like Vietnam, Malaysia, and Thailand, partially comprise the remaining production share, but Chinese companies often own this production.²⁸ In March 2023, India had 38 GW of production capacity for solar modules, comprising approximately 3 percent of the global production capacity.²⁹ Current Indian companies producing solar modules include Waaree, Adani Solar, Vikram Solar, Goldi, and RenewSys.³⁰ Notably, by 2025, India is projected to be the largest module producer outside of China.³¹ Yet, India largely relies on imports of solar cells and components from China to make its solar modules.³²

Figure 2: Solar PV Manufacturing Capacity by Country and Region in 2021



Author creation. Source: "Solar PV Manufacturing Capacity by Country and Region, 2021," International Energy Agency, updated July 5, 2022, <https://www.iea.org/data-and-statistics/charts/solar-pv-manufacturing-capacity-by-country-and-region-2021>. License CC BY 4.0.

²⁷ Dvorak and Mollica, "Can the US Break China's Grip on Solar?"

²⁸ IEA, "Special Report on Solar PV Global Supply Chains," 27.

²⁹ Gulia et al., "India's Photovoltaic Manufacturing Capacity," 10; "Solar PV Manufacturing Capacity by Component in India, 2021–2024," IEA.

³⁰ Gulia et al., "India's Photovoltaic Manufacturing Capacity," 12.

³¹ "China to Hold Over 80% of Global Solar Manufacturing Capacity from 2023–26," Wood Mackenzie, November 7, 2023, <https://www.woodmac.com/press-releases/china-dominance-on-global-solar-supply-chain/>.

³² IEA, "Special Report on Solar PV Global Supply Chains," 27.

3. INDIA'S COMPETITIVE ADVANTAGES AND DISADVANTAGES

India is well-positioned to become a global supplier of solar cells and especially solar modules given its relatively low labor costs and existing economies of scale, as well as increasing domestic and overseas demand for India-made solar cells and modules.³³ Domestically, India's installed solar PV capacity was 75.6 GW in February 2024, but India is targeting 300 GW of solar PV capacity by 2030, necessitating a significant build-out of solar projects.³⁴ This 2030 deployment target should provide reliable demand for domestically produced solar PV modules, especially with India's Approved List of Models and Manufacturers (ALMM) requiring government and government-assisted projects — which represent about 70 percent of India's module demand — to source India-made solar PV modules.³⁵ The ALMM order was reimposed on April 1, 2024, after it was suspended for the 2023–2024 fiscal year to allow Indian solar project developers to use cheaper Chinese solar modules in their projects and ostensibly to enable quicker deployment of solar PV projects.³⁶ Therefore, the ALMM should help increase demand for India-made solar modules and incentivize Indian module production.

India also has imposed a 40 percent basic customs duty on imported solar modules and a 25 percent basic customs duty on imported solar cells, which should have helped increase the cost-competitiveness of India-made cells and modules versus imported cells and modules.³⁷ However, solar cells and modules can enter India tariff-free from countries with whom India has free trade agreements, resulting in Chinese-made solar modules and cells entering India tariff-free through countries like Vietnam and Malaysia.³⁸

³³ Mondal, "India Wants Its Own Solar Industry."

³⁴ "Renewable Energy," Invest India, Government of India, accessed April 4, 2024, <https://www.investindia.gov.in/sector/renewable-energy>; "India Reaches 60 GW High-End Solar Module Manufacturing Capacity: AISMA," *Business Standard*, November 9, 2023, https://www.business-standard.com/industry/news/india-reaches-60-gw-high-end-solar-module-manufacturing-capacity-aisma-123110900683_1.html.

³⁵ Ministry of New and Renewable Energy, "Updation of List I (Manufacturers and Models of Solar PV Modules) of ALMM Order, 2019 – Reg.," August 3, 2023, <https://cdnbbsr.s3waas.gov.in/s3716e1b8c6cd17b771da77391355749f3/uploads/2023/08/2023080818.pdf>; Aarushi Koundal, "India's Solar Panel Imports Set to Remain Higher in FY24," *Economic Times*, December 25, 2023, <https://energy.economictimes.indiatimes.com/news/renewable/indias-solar-panel-imports-set-to-remain-higher-in-fy24/106217488>; Aarushi Koundal, "MNRE Brings Back ALMM for Solar Modules; to Be in Effect from 1 April 2024," *Economic Times*, February 12, 2024, <https://energy.economictimes.indiatimes.com/news/renewable/mnre-brings-back-alm-for-solar-modules-to-be-in-effect-from-1-april-2024/107622705>; "ALMM to Keep Margins of Solar Module Makers at 12–14% Next Fiscal," CRISIL Ratings, January 29, 2024, <https://www.crisilratings.com/en/home/newsroom/press-releases/2024/01/alm-to-keep-margins-of-solar-module-makers-at-12-14percent-next-fiscal.html>. As of April 8, 2024, India's Ministry of New and Renewable Energy has not yet released a list of approved solar cell producers for government-backed solar projects, but it will likely include some foreign producers. See Ministry of New and Renewable Energy, "Approved List of Models and Manufacturers (ALMM): Approved Models and Manufacturers of Solar Photovoltaic Modules (Requirements for Compulsory Registration) Order, 2019," accessed April 19, 2024, <https://mnre.gov.in/approved-list-of-models-and-manufacturers-alm/>; Koundal, "India's Solar Panel Imports Set to Remain Higher."

³⁶ Aarushi Koundal, "ALMM Should Be Kept in Abeyance for At Least One More Year: Industry," *Economic Times*, February 1, 2024, <https://energy.economictimes.indiatimes.com/news/renewable/alm-should-be-kept-in-abeyance-for-at-least-one-more-year-industry/107311555>; Satish Shetty, "ALMM Reimposed from April, No Exemptions for Open Access Solar Projects," *Mercom India*, March 29, 2024, <https://www.mercomindia.com/alm-reimposed-april-no-exemptions-open-access-solar>; Koundal, "India's Solar Panel Imports Set to Remain Higher"; Koundal, "MNRE Brings Back ALMM"; "India Could Become the World's Second-Largest Solar Photovoltaic Manufacturer by 2026," Institute for Energy Economics and Financial Analysis, April 4, 2023, <https://ieefa.org/articles/india-could-become-worlds-second-largest-solar-photovoltaic-manufacturer-2026>.

³⁷ Sudarshan Varadhan, "India to Levy Import Tax on Solar Modules, Cells from April 2022 — Memo," *Reuters*, March 9, 2021, <https://www.reuters.com/world/china/india-levy-import-tax-solar-modules-cells-april-2022-memo-2021-03-10/>.

³⁸ Vinod Mahanta, "Import of Cheaper Solar Modules on Rise As China Exploits Loopholes," *Economic Times*, December 7, 2023, <https://economictimes.indiatimes.com/industry/renewables/import-of-cheaper-solar-modules-on-rise-as-china-exploits-loopholes/articleshow/105104240.cms?from=mdr>.

While India's imports of solar modules are expected to increase from 2023 to 2024, the ALMM should help stem this trend by requiring government and government-assisted solar projects to source solar modules from India and solar cells from approved producers.³⁹

The export market is and will be a key source of demand for Indian solar PV cells and modules, particularly in case of limited Indian demand for India-made solar PV cells and modules due to lower-priced and higher-quality Chinese-produced products.⁴⁰ Indian exports of solar cells and modules have already been increasing, with solar cell and module exports increasing over 1,000 percent from the April–July 2022 timeframe versus the April–July 2023 timeframe.⁴¹ Just as China's solar PV production features significant exports because domestic production exceeds domestic demand, India's solar PV production could also heavily feature exports, especially to countries with high targets for installed solar PV modules, relatively low solar PV module manufacturing capacity, and a desire to diversify imports away from China.⁴²

India's solar PV exports will face stiff price competition due to the global overcapacity of solar PV products, largely caused by Chinese production.⁴³ For instance, Chinese solar PV modules cost \$0.11/watt (W) while Indian solar PV modules cost \$0.22/W.⁴⁴ However, Indian modules will still be price competitive in countries like Germany, where the average cost of installing residential solar is approximately \$1.70/Watt.⁴⁵ In other words, solar PV modules comprise a minority share of overall costs; thus, price discrepancies between solar PV modules from China and India have a correspondingly minor impact on overall costs. Some companies may also consider sourcing the more expensive Indian modules due to their lower supply chain and reputational risks versus sourcing modules from China. Indian solar PV exports could be especially price competitive in countries with import duties on solar PV cells and modules produced in China and other countries seek-

³⁹ Koundal, "India's Solar Panel Imports Set to Remain Higher,"; Koundal, "MNRE Brings Back ALMM."

⁴⁰ Gulia et al., "India's Photovoltaic Manufacturing Capacity," 7; Sudheer Singh and Aarushi Koundal, "Historic Dip in Chinese Solar Module Prices Set to Boost India's Solar Capacity Addition," *Economic Times*, July 11, 2023, <https://energy.economictimes.indiatimes.com/news/renewable/historic-dip-in-chinese-solar-module-prices-set-to-boost-indias-solar-capacity-addition/101657675>.

⁴¹ M. Ramesh, "Solar Module Exports Up 1000% As US Keeps China Out," *Hindu BusinessLine*, October 12, 2023, <https://www.thehindubusinessline.com/economy/solar-module-exports-up-1000-pc-as-us-keeps-china-out/article67408232.ece>.

⁴² IEA, "Special Report on Solar PV Global Supply Chains," 7.

⁴³ Andrew Hayley, "China Solar Industry Faces Shakeout, but Rock-Bottom Prices to Persist," *Reuters*, April 3, 2024, <https://www.reuters.com/business/energy/china-solar-industry-faces-shakeout-rock-bottom-prices-persist-2024-04-03/>; "Solar Factories Slow Output As Overcapacity Weighs on Profits," *Bloomberg*, January 25, 2024, <https://www.bloomberg.com/news/articles/2024-01-26/solar-factories-slow-output-as-overcapacity-weighs-on-profits>.

⁴⁴ Ed Crooks, "China's Solar Growth Sends Module Prices Plummeting," *Wood Mackenzie*, April 5, 2024, <https://www.woodmac.com/news/opinion/chinas-solar-growth-sends-module-prices-plummeting/>; "China Solar Panel Costs Drop 42% from Year Ago – Report," *Reuters*, December 14, 2023, <https://www.reuters.com/world/china/china-solar-panel-costs-drop-42-year-ago-report-2023-12-14/>.

⁴⁵ Michael Fuhs, "Germany's Average Residential PV Prices Rose by 10% to €1,557/kW in Q2," *pv magazine*, June 22, 2023, <https://www.pv-magazine.com/2023/06/22/germanys-average-residential-pv-prices-rose-by-10-to-e1557-kw-in-q2/>.

ing to circumvent such tariffs, namely Southeast Asian countries.⁴⁶

In the downstream supply chain like solar modules, India is competitive with China on the investment costs for solar module production too, yet India's higher operating costs due to higher energy costs and lower labor productivity make solar module manufacturing 9 percent more costly in India than in China.⁴⁷ These variable costs could decrease as Indian companies increase their economies of scale in solar-related manufacturing.⁴⁸ Importantly, India's relatively low costs mean India could become a key low-cost alternative supplier of solar PV modules for Western countries seeking to diversify their imports away from China. Indeed, Indian products comprised about 9 percent of the volume of U.S. solar imports in the first eleven months of 2023.⁴⁹

In the upstream supply chain like polysilicon, India has lower investment costs than Europe and the United States, positioning India as an alternative supplier to China in these products too.⁵⁰ India will, however, face greater challenges in becoming a global supplier of polysilicon, ingots, and wafers. These inputs are more capital-intensive, energy-intensive, and technology-intensive than solar cells and modules, resulting in greater capital expenditures, operating costs, and technical complexities.⁵¹ For example, new Chinese polysilicon plants with annual production capacities of 100,000 metric tons have average investment costs of \$1.2 billion.⁵² Globally, ingot and wafers plants have investment costs ranging from \$300 million to \$1.2 billion depending on their production capacities, although the Chinese company Trina Solar announced in 2023 that it would build a 25 GW, \$1.55 billion monocrystalline ingot factory in Sichuan province.⁵³ Similarly, the Chinese company JA Solar announced that it would invest \$870 million in an integrated solar project with an annual production capacity of 30

⁴⁶ For example, the United States has imposed duties on imported Chinese solar PV modules, helping reduce Chinese solar modules' cost advantages. The U.S. government has also imposed duties on Chinese solar PV producers that "finish" their products in Southeast Asian countries in order to circumvent US tariffs on Chinese solar products. Additionally, the U.S. government has seized thousands of solar-related shipments from China under the Uyghur Forced Labor Protection Act due to suspected ties to forced labor. See Lilly Yejin Lee and Noah Kaufman, "Q&A | Solar Tariffs and the US Energy Transition," Center on Global Energy Policy, Columbia University, November 13, 2023, <https://www.energypolicy.columbia.edu/qa-solar-tariffs-and-the-us-energy-transition/>; Nichola Groom, "US Places Tariffs on Some Big Solar Companies for Dodging China Duties," *Reuters*, August 18, 2023, <https://www.reuters.com/sustainability/us-slaps-tariffs-some-big-solar-companies-dodging-china-duties-2023-08-18/>; Andrea Day and Chris DiLella, "Since June, Feds Have Seized Nearly \$1 Billion in Goods Tied to Forced Labor," *CNBC*, April 17, 2023, <https://www.cnn.com/2023/04/17/feds-seize-goods-tied-to-forced-labor.html>; Nichola Groom, "Exclusive: US Blocks More than 1,000 Solar Shipments over Chinese Slave Labor Concerns," *Reuters*, November 11, 2022, <https://www.reuters.com/world/china/exclusive-us-blocks-more-than-1000-solar-shipments-over-chinese-slave-labor-2022-11-11/>.

⁴⁷ IEA, "Special Report on Solar PV Global Supply Chains," 86–87. Labor costs per hour are lower in India than China, but labor productivity is lower in India than China as Chinese factories are more automated. Consequently, overall labor costs are higher in India than China.

⁴⁸ IEA, "Special Report on Solar PV Global Supply Chains," 11.

⁴⁹ Sheridan Prasso, "US Solar Boom Opens \$2 Billion Indian Door to Banned Products from China," *Bloomberg*, February 5, 2024, <https://www.bloomberg.com/news/articles/2024-02-05/us-solar-boom-opens-indian-door-to-chinese-components-made-with-forced-labor>; d Ramesh, "Solar Module Exports Up 1000%."

⁵⁰ IEA, "Special Report on Solar PV Global Supply Chains," 85.

⁵¹ Gulia et al., "India's Photovoltaic Manufacturing Capacity," 10; Mondal, "India Wants Its Own Solar Industry."

⁵² Johannes Bernreuter, "Why the Pork Cycle Has Returned to the Polysilicon Market," Bernreuter Research, January 16, 2024, <https://www.bernreuter.com/newsroom/polysilicon-news/article/why-the-pork-cycle-has-returned-to-the-polysilicon-market/>.

⁵³ IEA, "Special Report on Solar PV Global Supply Chains," 85; Vincent Shaw, "Trina to Build 25 GW Ingot Factory, Longi Reduces Wafer Prices by 30%," *PV Magazine*, June 2, 2023, <https://www.pv-magazine-india.com/2023/06/02/trina-to-build-25-gw-ingot-factory-longi-reduces-wafer-prices-by-30/>.

GW of ingots, 10 GW of wafers, and 10 GW of modules.⁵⁴

Compounding the challenges of capital intensity, solar PV projects face long lead times in commissioning and ramping up production, which delays revenue generation. Estimated lead times from project announcement through to commissioning for polysilicon plants in India are approximately 18 to 30 months, and these plants also have a long ramp-up until they meet nameplate capacity.⁵⁵ Downstream projects have less lead time, meaning they generate revenue more quickly and pose less investment risks. Lead times for ingot and wafer plants in India are 8 to 20 months, while lead times for cell and module plants in India are 5 to 11 months.⁵⁶

4. VERTICAL INTEGRATION

Despite these upstream challenges, India should still seek to produce polysilicon, ingots, and wafers as it will enable vertical integration and reduce India's import reliance on China, better positioning India as a true alternative supplier to China. Already, the Indian companies Indosol Solar (a special purpose vehicle of Shirdi Sai Electricals), Reliance New Energy Solar, Adani Infrastructure, and FS India Solar Ventures intend to build integrated solar PV production facilities, which encompass production from the polysilicon to the modules.⁵⁷ Companies with vertically integrated facilities in the solar PV supply chains increase their price competitiveness because it decreases variable costs, such as transportation costs.⁵⁸ Consequently, vertically integrated companies are more profitable than pure-play companies that focus on one supply chain segment.⁵⁹ For example, Canadian Solar, a Chinese company, is in the world's top ten companies for the largest market shares in wafers *and* cells *and* modules.⁶⁰ Therefore, vertically integrated Indian solar companies should be more competitive globally.

Indian vertical integration, especially the production of polysilicon, ingots, and wafers, will help reduce reliance on Chinese companies too. For example, the majority of Indian companies awarded PLI incentives rely on Chinese equipment suppliers.⁶¹ To illustrate, the Indian solar company Waaree listed seventeen supply chain partners, and all seventeen partners are Chinese companies.⁶² To further reduce reliance on China,

⁵⁴ Carrie Xiao, "JA Solar Invests RMB6.02 Billion to Expand Production Capacity in China," *PV Tech*, June 9, 2023, <https://www.pv-tech.org/ja-solar-invests-rmb6-02-billion-to-expand-production-capacity-in-china/>.

⁵⁵ IEA, "Special Report on Solar PV Global Supply Chains," 65–66.

⁵⁶ IEA, "Special Report on Solar PV Global Supply Chains," 66.

⁵⁷ "List of Successful Bidders under Tranche-I of PLI Scheme," Ministry of New and Renewable Energy; "Capacity Allocation under RfS for Selection of Solar PV Module Manufacturers," Ministry of New and Renewable Energy.

⁵⁸ IEA, "Special Report on Solar PV Global Supply Chains," 7, 88.

⁵⁹ IEA, "Special Report on Solar PV Global Supply Chains," 48.

⁶⁰ Jessica Jin, "Solar PV Module Supply Chain Accelerates Consolidation in the Top 10 Suppliers," S&P Global, February 14, 2024, <https://www.spglobal.com/commodityinsights/en/ci/research-analysis/solar-pv-module-supply-chain-accelerates-consolidation.html>.

⁶¹ "Production Linked Incentive (PLI) Scheme," Ministry of New and Renewable Energy. For the successful bidders under the PLI schemes, see "List of Successful Bidders under Tranche-I of PLI Scheme," Ministry of New and Renewable Energy; "Capacity Allocation under RfS for Selection of Solar PV Module Manufacturers," Ministry of New and Renewable Energy; "Chinese Vendors Dominate Supply Chain of Indian Solar PLI Scheme Winners: Report," *Mint*, February 8, 2024, <https://www.livemint.com/companies/news/chinese-vendors-dominate-supply-chain-of-indian-solar-pli-scheme-winners-report-11707365855802.html>.

⁶² *Mint*, "Chinese Vendors Dominate Supply Chain of Indian Solar PLI Scheme Winners."

India should also seek to develop production capacity in metallurgical-grade silicon, which is created from silica minerals. India produced no metallurgical-grade silicon in 2023, yet without the production of its own metallurgical-grade silicon, Indian polysilicon producers likely are forced to rely on Chinese suppliers: China produced 79 percent of the world's metallurgical-grade silicon in 2023.⁶³ In 2021, India produced 12 million metric tons of silica minerals, but India's high energy costs currently make the production of metallurgical-grade silicon "uneconomical" in India.⁶⁴

5. INTERNATIONAL COOPERATION

International cooperation, like concessional financing, could help Indian companies develop upstream production capacity in polysilicon, ingots, and wafers, and international cooperation could also help finance renewable energy projects, which could power solar-related manufacturing projects and reduce their electricity costs. As previously noted, the most challenging segments for vertical integration are the production of polysilicon, ingots, and wafers due to their high capital expenditures and high electricity demands.⁶⁵ However, high industrial electricity prices, rather than capital needs, pose a greater challenge in expanding such production capacity because Indian companies like Adani Solar have substantial capital, especially when paired with government incentives, for building production facilities.⁶⁶ Therefore, with international partners like the United States International Development Finance Corporation (DFC), India could collaborate on building renewable energy facilities that support production in the solar PV supply chain, such as polysilicon, ingots, and wafers. Already, the DFC has lent nearly \$1 billion for solar cell and module production in India.⁶⁷

Renewable energy could also lower the carbon emissions and electricity costs of Indian manufacturing in the solar PV supply chain. Solar modules only need to operate for four to eight months to offset their supply chain emissions, but a wholly Indian solar PV supply chain in the short term would be an estimated 15 percent more emis-

⁶³ Murphy and Elimä, "In Broad Daylight," 16; Emily K. Schnebele, "Silicon," U.S. Geological Survey, January 2024, 161, <https://pubs.usgs.gov/periodicals/mcs2024/mcs2024-silicon.pdf>. The U.S. Geological Survey refers to metallurgical-grade silicon as silicon metal.²⁸ IEA, "Special Report on Solar PV Global Supply Chains," 27.

⁶⁴ Robert C. Goodin, "Sand and Gravel (Industrial)," U.S. Geological Survey, January 2024, 155, <https://pubs.usgs.gov/periodicals/mcs2024/mcs2024-sand-industrial.pdf>; Paul Basore and David Feldman, "Solar Photovoltaics: Supply Chain Deep Dive Assessment," Solar Energy Technologies Office, U.S. Department of Energy, February 24, 2022, 13, <https://www.energy.gov/sites/default/files/2022-02/Solar%20Energy%20Supply%20Chain%20Report%20-%20Final.pdf>.

⁶⁵ IEA, "Special Report on Solar PV Global Supply Chains," 119.

⁶⁶ Gulia et al., "India's Photovoltaic Manufacturing Capacity," 17.

⁶⁷ "DFC Board of Directors Approves USD 425 Million in Financing for Tata Power's Greenfield 4.3 GW Solar Cell and Module Manufacturing Plant in Tamil Nadu," Tata Power, September 11, 2023, <https://www.tatapower.com/media/PressReleaseDetails/2054/dfc-board-of-directors-approves-usd-425-million-in-financing-for-tata-powers-greenfield-4-3-gw-solar-cell-and-module-manufacturing-plant-in-tamil-nadu>; "DFC Provides \$500 Million Loan to First Solar to Set Up 3.3 GW Manufacturing Facility in TN," *Economic Times*, January 11, 2024, <https://economictimes.indiatimes.com/industry/renewables/dfc-provides-500-million-loan-to-first-solar-to-set-up-3-3-gw-manufacturing-facility-in-tn/articleshow/106737154.cms?from=mdr>. The loans included \$425 million to TP Solar Limited, a wholly owned subsidiary of Tata Power Renewable Energy Limited, to build a 4 GW solar cell and module manufacturing facility in Tamil Nadu, and \$500 million to First Solar to build a 3.3 GW integrated solar manufacturing in Tamil Nadu. The US Development Finance Corporation has also signed a retainer letter for \$200 million to Vikram Solar for a 3 GW cell and module manufacturing facility. See "DFC and Vikram Solar Sign Ceremonial Retainer Letter for ~ \$200 Million Funding at COP28," U.S. Development Finance Corporation, December 15, 2023, <https://www.dfc.gov/media/press-releases/dfc-and-vikram-solar-sign-ceremonial-retainer-letter-200-million-funding-cop28>.

sions-intensive than in China given India's high reliance on coal.⁶⁸ However, India has a largely open-access regime with its power grid, which means solar-related manufacturers can opt to buy renewable energy directly from the grid (or build their own renewable energy generation) and lower their carbon emissions.⁶⁹ For instance, First Solar's factory in India will mainly use wind and solar electricity, despite India's overall grid relying primarily on fossil fuels.⁷⁰ With lower carbon sources of electricity like international-backed solar, wind, and storage projects, India's solar PV supply chains could lower their emissions intensity.⁷¹ Ultimately, cheaper renewable electricity could enable India to produce cheaper solar-related components with lower carbon intensity, especially in the electricity-intensive upstream segments of the solar PV supply chain.

This international cooperation could specifically target Indian states with the best conditions to support projects in the solar PV supply chain. Gujarat, for example, represents a high-potential location for vertically integrated solar PV production and renewable energy projects.⁷² It is the destination for almost 57 percent of India's future solar module capacity, and its low renewable electricity prices, high grid reliability, and easy port access make it an attractive location for solar-related projects, including for upstream supply segments like polysilicon.⁷³ Indeed, Adani Solar's production of India's first monocrystalline silicon ingot occurred at its Mundra plant in Gujarat.⁷⁴ Other Indian states with existing solar cell and module production as well as strong renewable resources are well-positioned for vertical integration with upstream production, including Telangana, Rajasthan, and Tamil Nadu.⁷⁵

6. CONCLUSION

India has existing production and latent potential to serve as an alternative supplier

⁶⁸ MIEA, "Special Report on Solar PV Global Supply Chains," 8, 78–80.

⁶⁹ Ammu Susanna Jacob and Rishu Garg, "Green Energy Open Access: Empowering Consumers with Clean Electricity," *Economic Times*, March 8, 2024, <https://energy.economicstimes.indiatimes.com/news/renewable/green-energy-open-access-empowering-consumers-with-clean-electricity/108315350>.

⁷⁰ Uma Gupta, "First Solar to Power Indian Module Factory with Wind, Solar under PPA," *pV magazine*, January 4, 2024, <https://www.pv-magazine.com/2024/01/04/first-solar-powering-indian-module-factory-with-wind-solar-under-ppa/>; Ministry of Power, "Power Sector at a Glance: All India," Government of India, updated June 12, 2023, <https://powermin.gov.in/en/content/power-sector-glance-all-india>.

⁷¹ G. J. Simandl, S. Paradi, and L. Simandl, "Future of Photovoltaic Materials with Emphasis on Resource Availability, Economic Geology, Criticality, and Market Size/Growth," *CIM Journal* 14, no. 3, (2023): 139, <https://doi.org/10.1080/19236026.2023.2168419>. India's National Institute of Solar Energy estimated that India's solar power potential is 748 GW. See "Solar Overview," Ministry of New and Renewable Energy, Government of India, accessed April 6, 2024, <https://mnre.gov.in/solar-overview>.

⁷² Arjun Joshi, "India's Total Solar Module Manufacturing Capacity 64.5 GW in 2023," Mercom India, March 28, 2024, <https://www.mercomindia.com/india-solar-module-manufacturing-2023/>; "State of Solar PV Manufacturing in India," Mercom India, 2024, accessed April 30, 2024, <https://mercomindia.s3.ap-south-1.amazonaws.com/wp-content/uploads/2024/03/StateofSolarPVManufacturinginIndia2024ExecSummary.pdf>.

⁷³ Gulia et al., "India's Photovoltaic Manufacturing Capacity," 13. Integrated solar PV production facilities can be designed for high water-stressed areas. For example, First Solar is building a new solar PV manufacturing facility in Tamil Nadu, an Indian state with high water stress. Consequently, "the facility has been designed to be a net-zero water withdrawal PV manufacturing facility, believed to be the world's first, and will rely entirely on tertiary treated reverse osmosis water from the city's sewage treatment plant and have zero wastewater discharge." See First Solar, "Sustainability Report 2023," 2023, 5, https://www.firstsolar.com/-/media/First-Solar/Sustainability-Documents/FirstSolar_Sustainability-Report_2023.ashx.

⁷⁴ "Adani Solar's Green Breakthrough," Adani Solar.

⁷⁵ Joshi, "India's Total Solar Module Manufacturing Capacity,"; "State of Solar PV Manufacturing in India," Mercom India.

to China in the solar PV supply chain, especially for solar cells and modules. It will, however, face greater challenges in becoming a global supplier of polysilicon, ingots, and wafers, which are more capital-intensive, energy-intensive, and technology-intensive than solar cells and modules. Nonetheless, Indian companies should still seek to produce polysilicon, ingots, and wafers as it will facilitate vertical integration and reduce import reliance on China. International cooperation, like concessional financing, could help Indian companies develop upstream production capacity in polysilicon, ingots, and wafers, and international cooperation could also help finance renewable energy projects, which could power solar-related manufacturing projects and reduce their electricity costs.

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CHARGING UP: INDIA'S POTENTIAL ROLE IN GLOBAL BATTERY SUPPLY CHAINS

BY GREGORY WISCHER

SUMMARY

India could help advance the global electric vehicle transition by producing certain goods in segments of the global battery supply chain. Currently, India lacks a sizable production share in any segment of the global battery supply chain, which is largely dominated by China. India does have existing production and significant growth potential in specific segments of the battery supply chain, especially as some countries seek to diversify their supply chains away from China due to geopolitical concerns. For instance, India already produces several of the ancillary raw and precursor materials needed for battery manufacturing, but it uses them for other goods. Yet, India faces other challenges such as limited resources of lithium, nickel, and cobalt. To incorporate India into the global battery supply chain, India and the international community should collaborate on trade, investment and financing, and research.

1. INTRODUCTION

The global automotive industry is experiencing a major transformation. Vehicles fully powered by electric batteries comprise an increasing share of new vehicle purchases.¹ Compared to internal combustion engine vehicles, battery electric vehicles run fully on electricity, not gasoline or diesel. Therefore, battery electric vehicles do not directly emit carbon dioxide, and if the batteries' electricity source has low or no carbon dioxide emissions, increased electric vehicle adoption can lower carbon dioxide emissions in the transportation sector, helping countries meet climate-related goals. For example, in India, the carbon dioxide emissions of electric vehicles are 8 percent to 24 percent lower than internal combustion engine vehicles over their lifetimes.² Transitioning to electric vehicles can also strengthen countries' energy security by reducing import dependence on oil, and it can increase countries' economic prosperity by creating new market opportunities. Thus, many countries, including India, are incentivizing electric vehicle adoption.³

However, the global manufacturing supply chain for the core part of electric vehicles — their batteries — is dominated by the People's Republic of China.⁴ Given concerns of overdependence on China, many countries are seeking to shift their battery supply chains to other countries, including India.⁵ Across all segments of the battery supply chain, India's production is presently negligible, but Indian companies have existing mineral

¹ Securing America's Future Energy, "The Commanding Heights of Global Transportation," September 15, 2020, 1, <https://safe2020.wpenginepowered.com/wp-content/uploads/2020/09/The-Commanding-Heights-of-Global-Transportation-1.pdf>; Brian Kahn, "The World Hit 'Peak' Gas-Powered Vehicle Sales — in 2017," *Bloomberg*, January 30, 2024, <https://www.bloomberg.com/news/articles/2024-01-30/world-hit-peak-gas-powered-vehicles-as-evs-gain-market-share>.

² Abhinav Soman, Karthik Ganesan, and Harsimran Kaur, "India's Electric Vehicle Transition: Impact on Auto Industry and Building the EV Ecosystem," Council on Energy, Environment and Water, October 2019, 30, <https://www.ceew.in/sites/default/files/CEEW-IndiaElectricVehicleTransitionReportPDF26Nov19.pdf>.

³ International Energy Agency, "Global EV Outlook 2023: Catching Up with Climate Ambitions," April 2023, 63–89, <https://iea.blob.core.windows.net/assets/dacf14d2-eabc-498a-8263-9f97fd5dc327/GEVO2023.pdf>.

⁴ The electric vehicle battery constitutes between 30 percent and 40 percent of the vehicle's overall cost. See Adrian König et al., "An Overview of Parameter and Cost for Battery Electric Vehicles," *World Electric Vehicle Journal* 12, no. 1 (2021): 2, <https://doi.org/10.3390/wevj12010021>; Institute for Energy Research, "Electric Vehicle Battery Costs Soar," April 25, 2022, <https://www.instituteforenergyresearch.org/renewable/electric-vehicle-battery-costs-soar/>; Rahul Gupta et al., "The Real Global EV Buzz Comes on Two Wheels," McKinsey & Company, August 2023, 6, https://www.mckinsey.com/-/media/mckinsey/industries/automotive%20and%20assembly/our%20insights/the%20real%20global%20ev%20buzz%20comes%20on%20two%20wheels/the-real-global-ev-buzz-comes-on-two-wheels_final-v4.pdf?shouldIndex=false.

⁵ Joe Cash, "Western Firms Shift Investment from China to India As Worries Mount," *Reuters*, September 13, 2023, <https://www.reuters.com/markets/western-firms-shift-investment-china-india-worries-mount-2023-09-13/>.

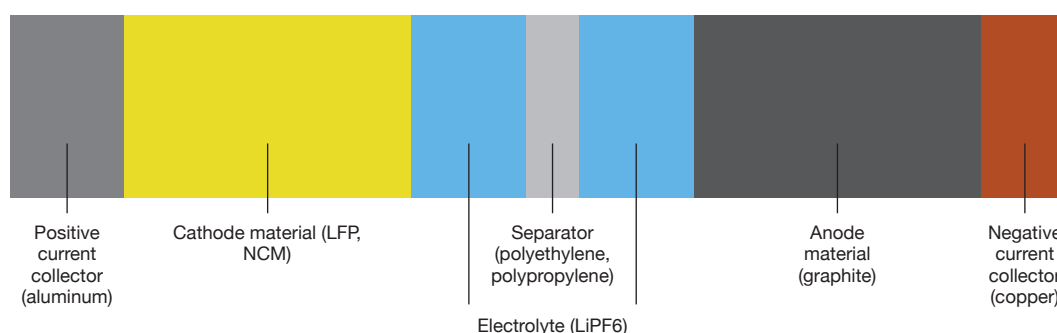
production, processing expertise, battery cell investments, battery pack assembly capacity, and recycling experience. Specifically, India has the greatest production potential in certain raw materials, precursor materials, lithium-iron-phosphate (LFP) battery cells, battery packs for two-wheeled (scooters, motorbikes) and three-wheeled vehicles (rickshaws), and black mass (shredded end-of-life batteries). Consequently, India could help advance the global electric vehicle transition by producing certain goods in segments of the global battery supply chain.

This analysis seeks to contextualize India's present and potential role in the global supply chain for electric vehicle batteries. Considering India's production potential in the battery supply chain, the paper concludes by recommending that India and the international community collaborate on trade, investment and financing, and research in the battery supply chain. Such efforts would further incorporate India into the global supply chain for electric vehicle batteries and thus support the global transition to electric vehicles.

2. THE GLOBAL BATTERY SUPPLY CHAIN AND CHINA'S DOMINATION

The critical part of a battery electric vehicle is the battery pack, which contains individual battery cells made with various materials. The cell contains the following six key components: cathode material, anode material, electrolyte, separator, positive current collector, and negative current collector (Figure 1). The cathode is the most costly part of the battery, representing between 50 percent and 60 percent of battery cell costs, and 90 percent of the cathode material costs are raw materials, which depend on the battery chemistry.⁶ The most common cathode chemistries today are nickel-cobalt-manganese (NCM) cathodes and LFP cathodes. The other core part of the battery is the anode, which is mainly graphite. Ultimately, these cell components function together to convert chemical energy into electricity that powers the vehicle.

Figure 1: The Battery Cell - Building Block of a Battery Pack



A battery pack contains multiple components made of various raw materials.

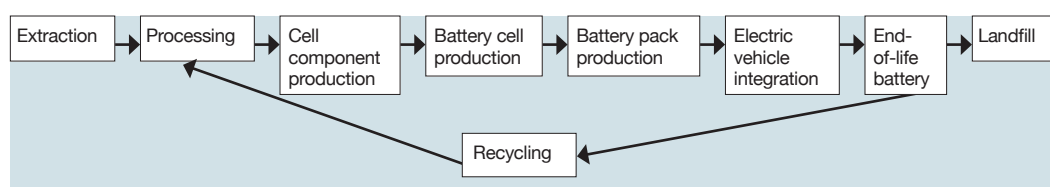
Source: Author schematic adapted from Dhruv Warrior, Akanksha Tyagi, and Rishabh Jain, "How Can India Indigenise Lithium-Ion Battery Manufacturing? Formulating Strategies across the Value Chain," Council on Energy, Environment and Water, February 2023, 10, <https://www.ceeew.in/sites/default/files/how-can-india-scale-lithium-ion-battery-manufacturing-sector-and-supply-chain.pdf>. Copyright © 2023 Council on Energy, Environment and Water (CEEW). This work is licensed under the Creative Commons Attribution-NonCommercial 4.0. International (CC BY-NC 4.0) license.

⁶ Ahmed Mehdi and Tom Moerenhout, "The IRA and the US Battery Supply Chain: Background and Key Drivers," Center on Global Energy Policy, Columbia University, June 2023, 5, https://www.energypolicy.columbia.edu/wp-content/uploads/2023/06/BatteryValueChain_Commentary_CGEP.pdf.

The manufacturing supply chain for electric vehicle batteries proceeds from raw materials and precursor materials to cell components and battery cells and packs, eventually reaching the end of their useful lives when some batteries are recycled and reincorporated into the battery supply chain. The supply chain for producing electric vehicle batteries can be illustrated by the following steps (Figure 2):

1. Extraction: recovering minerals (like lithium, nickel, cobalt, manganese, and graphite) from a resource, such as hard rock ore.
2. Processing: refining minerals into precursor materials (for example, transforming natural graphite into spherical graphite).
3. Cell component production: producing the electrolyte, separator, current collectors, cathode material, and anode material (such as converting spherical graphite into anode material).
4. Battery cell production: combining the cell components into battery cells.
5. Battery pack production: assembling the battery cells into battery packs for electric vehicles.
6. Electric vehicle integration: integrating the battery packs with the electric vehicles.
7. End-of-life battery: deciding whether to dispose of the end-of-life battery or recycle it.
8. Recycling: re-using or shredding end-of-life batteries into material (black mass) that can be reprocessed into precursor materials.

Figure 2: The Battery Supply Chain in Phases

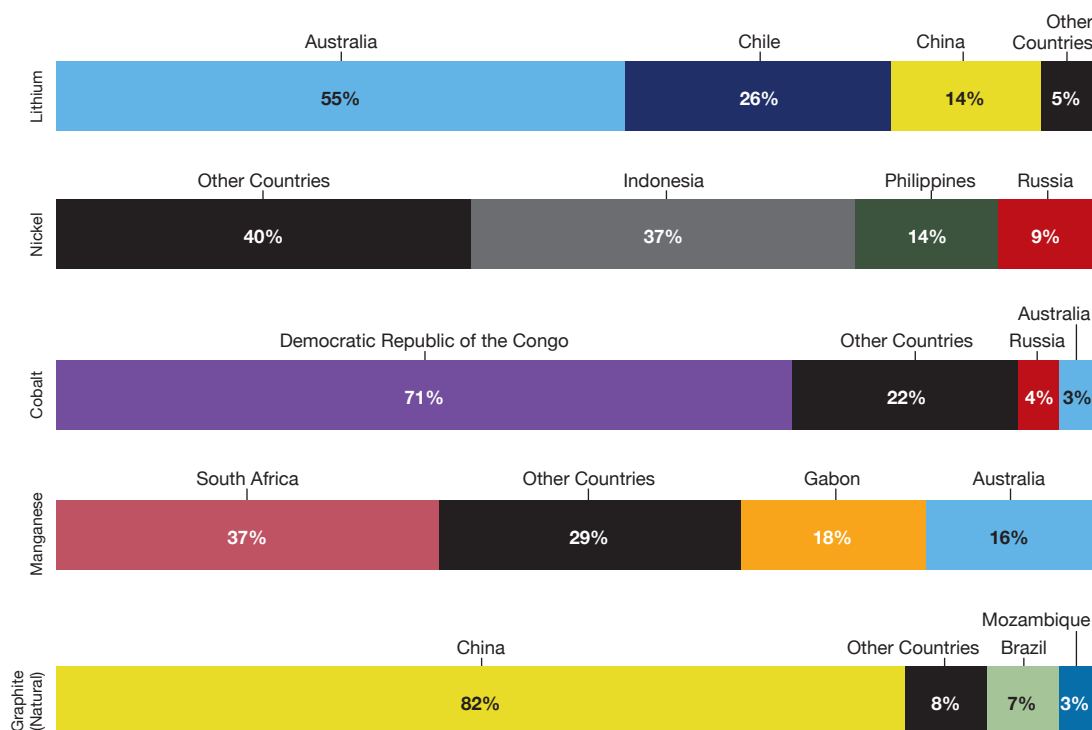


Source: Author creation adapted from International Energy Agency, “Global Supply Chains of EV Batteries,” July 2022, 19, <https://iea.blob.core.windows.net/assets/4eb8c252-76b1-4710-8f5e-867e751c8dda/GlobalSupplyChainsofEVBatteries.pdf>.

As noted, batteries contain significant amounts of raw materials, which are extracted globally (Figure 3). Yet, the companies extracting these raw materials are often from China. For instance, Chinese companies owned between 40 percent and 50 percent of cobalt production in the Democratic Republic of the Congo in 2020.⁷ Major non-Chinese mining companies such as BHP and Rio Tinto have generally focused on producing raw materials with larger volumes, like iron ore and aluminum, but they are increasing their mining activities in battery-related raw materials, like nickel and lithium.⁸

⁷ Luiza Ch. Savage, “How America Got Outmaneuvered in a Critical Mining Race,” *Politico*, December 2, 2020, <https://www.politico.com/news/2020/12/02/china-cobalt-mining-441967>.

⁸ Fumi Matsumoto, “BHP Bets Big on EV-Driven Nickel boom,” *Nikkei Asia*, August 25, 2023, <https://asia.nikkei.com/Business/Materials/BHP-bets-big-on-EV-driven-nickel-boom>; BloombergNEF, “Rio Tinto Doubles Down on Lithium Mining with Rwanda Deal,” *Bloomberg*, September 22, 2023, <https://www.bloomberg.com/professional/blog/rio-tinto-doubles-down-on-lithium-mining-with-rwanda-deal/>. Illustrating the difference in annual production volume for these materials, the world in 2023 mined an estimated 1.5 billion metric tons of iron in iron ore and 22 million metric tons of copper, versus 3.6 million metric tons of nickel and 180,000 metric tons of lithium. See Candice Tuck, “Iron Ore,” US Geological Survey, January 2024, 101, <https://pubs.usgs.gov/periodicals/mcs2024/mcs2024-iron-ore.pdf>; Daniel M. Flanagan, “Copper,” US Geological Survey, January 2024, 65, <https://pubs.usgs.gov/periodicals/mcs2024/mcs2024-copper.pdf>; Michele E. McRae, “Nickel,” US Geological Survey, January 2024, 125, <https://pubs.usgs.gov/periodicals/mcs2024/mcs2024-nickel.pdf>; Brian W. Jaskula, “Lithium,” US Geological Survey, January 2024, 111, <https://pubs.usgs.gov/periodicals/mcs2024/mcs2024-lithium.pdf>.

Figure 3: Estimated Share of Global Extraction for Minerals Common in Batteries in 2021

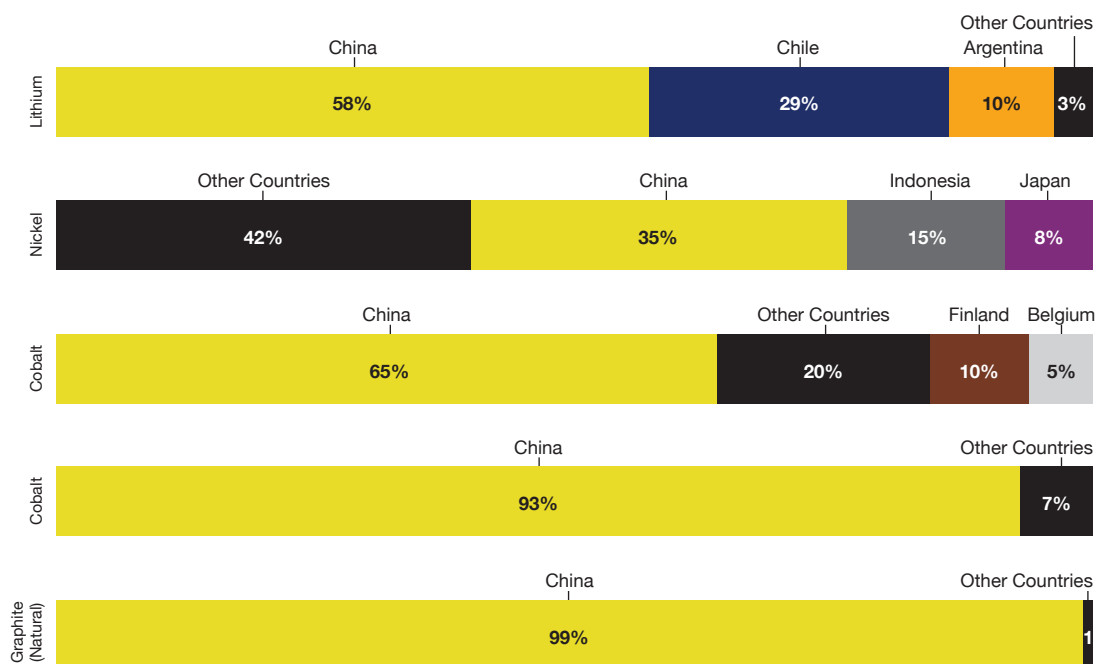
Source: Author creation based on data from Organisation for Economic Co-operation and Development, "Strengthening Clean Energy Supply Chains for Decarbonisation and Economic Security: OECD Report for the G7 Finance Ministers and Central Bank Governors," May 2023, 4, <https://www.oecd.org/trade/topics/global-value-chains-and-trade/documents/clean-energy-supply-chains-OECD-G7-202305.pdf>.

In addition to controlling significant raw materials extraction overseas, Chinese companies dominate the downstream segments of the battery supply chain. China comprises a majority of global production for most precursor materials (see Figure 4), and it produces a majority of the world's cell components, including separators, electrolyte salt, cathode material, and anode material (Figure 5).⁹ China is also the leading producer of batteries, with major battery companies like Contemporary Amperex Technology Co., Limited (CATL), Build Your Dreams (BYD), and Guoxuan High-Tech (Figure 5).¹⁰ China is the world's leading battery recycler too, but the global battery recycling segment is nascent with many governments yet to adopt extensive regulations.¹¹

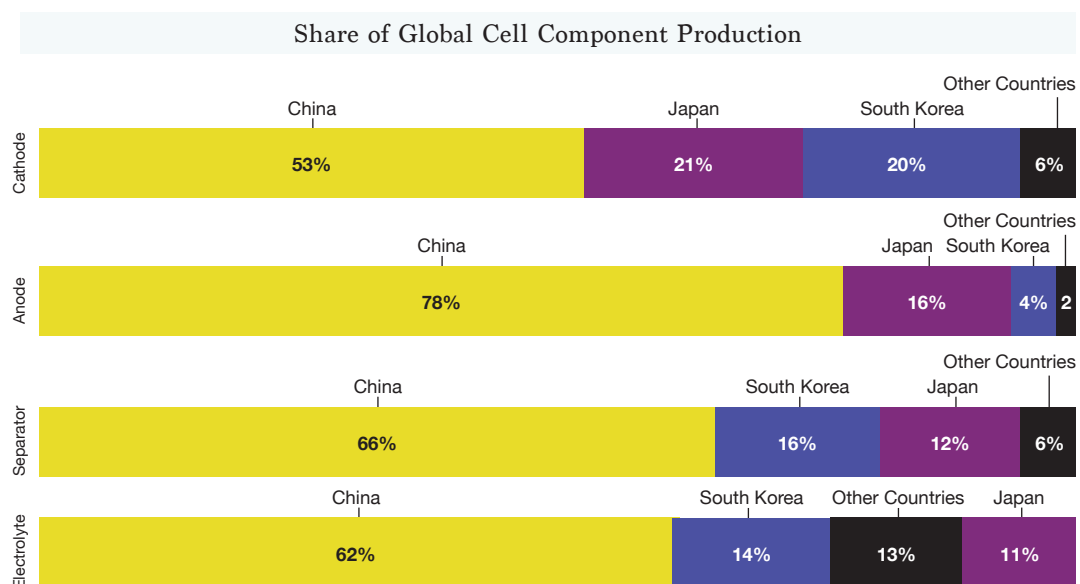
⁹ Organisation for Economic Co-operation and Development, "Strengthening Clean Energy Supply Chains for Decarbonisation and Economic Security: OECD Report for the G7 Finance Ministers and Central Bank Governors," May 2023, 4–5, <https://www.oecd.org/trade/topics/global-value-chains-and-trade/documents/clean-energy-supply-chains-OECD-G7-202305.pdf>; Abdurrehman Naveed and Cina Vazir, "Value amidst Transition: Evaluating Strategic Opportunities for Value Addition in the Democratic Republic of Congo," Mossavar-Rahmani Center for Business & Government, Harvard University, June 2023, 59–60, 63, https://www.hks.harvard.edu/sites/default/files/centers/mrcbg/working.papers/204_AWP_final.pdf.

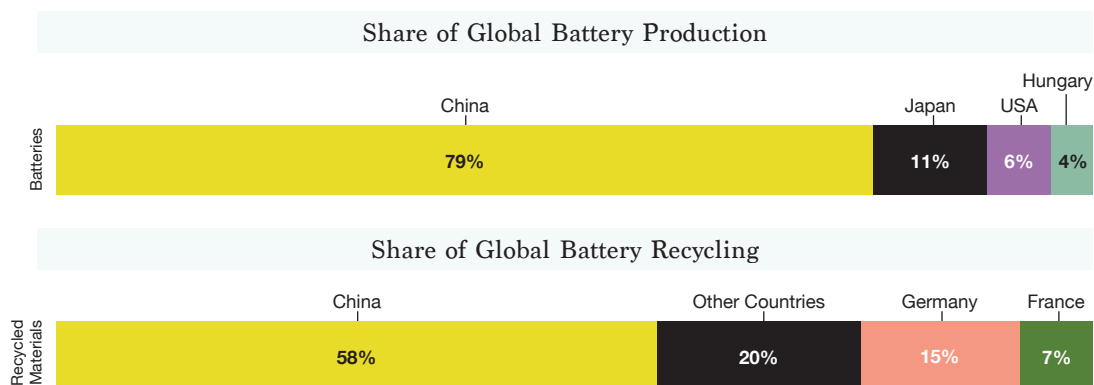
¹⁰ Organisation for Economic Co-operation and Development, "Strengthening Clean Energy Supply Chains for Decarbonisation and Economic Security," 5; and SNE Research, "From Jan to Sep in 2023, Global EV Battery Usage Posted 485.9GWh, a 44.4% YoY Growth," November 7, 2023, https://www.sneresearch.com/en/insight/release_view/183/page/0?s_cat=%7C&s_keyword=.

¹¹ Organisation for Economic Co-operation and Development, "Strengthening Clean Energy Supply Chains for Decarbonisation and Economic Security," 5; "Flood of Used Batteries Could Supercharge China's Electric Car Market," *Bloomberg*, December 20, 2023, <https://www.bloomberg.com/news/features/2023-12-20/recycling-spent-batteries-is-china-s-next-big-electric-car-challenge>.

Figure 4: Share of Global Processing for Minerals Common in Batteries in 2019


Source: Author creation based on data from Organisation for Economic Co-operation and Development, "Strengthening Clean Energy Supply Chains for Decarbonisation and Economic Security: OECD Report for the G7 Finance Ministers and Central Bank Governors," May 2023, 4-5, <https://www.oecd.org/trade/topics/global-value-chains-and-trade/documents/clean-energy-supply-chains-OECD-G7-202305.pdf>.

Figure 5: Share of Global Cell Component Production, Battery Production, and Recycling in 2020




Countries' share of global battery production combines both cell and pack production.

Source: Author creation based on data from Organisation for Economic Co-operation and Development, "Strengthening Clean Energy Supply Chains for Decarbonisation and Economic Security: OECD Report for the G7 Finance Ministers and Central Bank Governors," May 2023, 5, <https://www.oecd.org/trade/topics/global-value-chains-and-trade/documents/clean-energy-supply-chains-OECD-G7-202305.pdf>.

3. INDIA'S POTENTIAL ROLE IN GLOBAL BATTERY SUPPLY CHAINS

Across the battery supply chain, India lacks notable production capacity, but it has existing production and significant growth potential in certain goods. For raw materials, India does not produce lithium, nickel, and cobalt, yet it produces other raw materials necessary in the battery supply chain like copper, graphite, and manganese.¹² For precursor materials, India lacks processing capacity for several precursor materials such as lithium carbonate, but it has expertise producing other precursor materials such as aluminum, refined copper, and phosphoric acid. India also currently lacks production capacity in most cell components, yet Indian companies are building production capacity in cell components like anode material.

Similarly, India does not have sizable production capacity for battery cells (i.e., less than 1 percent of global capacity), but Indian companies are building battery cell production facilities, with LFP chemistries estimated to represent 70 percent of India's future battery production.¹³ Furthermore, India already assembles battery packs for different types of electric vehicles, and with high downstream demand for two-wheeled and three-wheeled electric vehicles and government subsidies for purchasing such vehicles, India has considerable

¹² Ministry of Mines, "Annual Report 2022–23," Government of India, March 10, 2023, 250, <https://mines.gov.in/admin/storage/app/uploads/6433da09a9f741681119753.pdf>. For nickel, for example, the *Indian Minerals Yearbook 2022* states, "Nickel is not produced from primary sources in the country and the entire demand is met through imports." See "Nickel," in *Indian Minerals Yearbook 2022* (Nagpur: Indian Bureau of Mines, November 2023), 12-2, https://ibm.gov.in/writereaddata/files/170988010865eab32c0db25Nickel_2022.pdf.

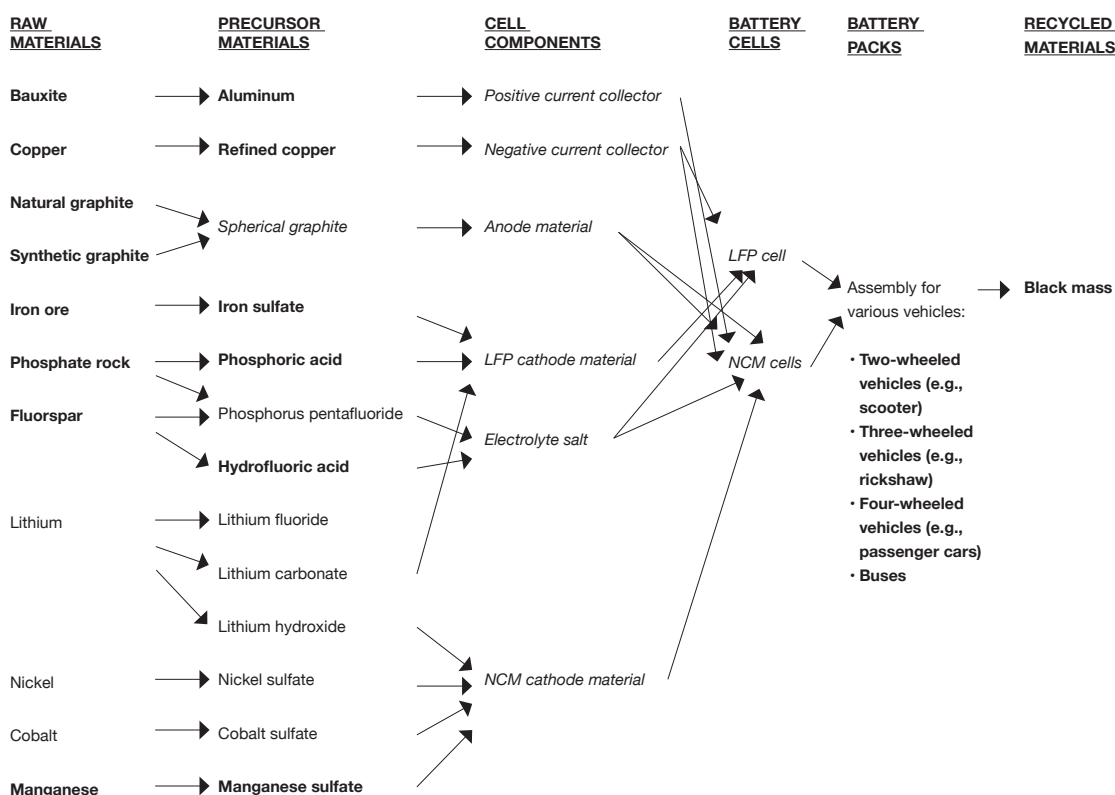
¹³ As of May 31, 2023, India had lithium-ion battery production capacity of 18 GWh, which is 0.018 TWh, while the global lithium-ion battery production capacity was 2.8 TWh. See Aude Marjolin, "Lithium-Ion Battery Capacity to Grow Steadily to 2030," S&P Global, July 27, 2023, <https://www.spglobal.com/marketintelligence/en/news-insights/research/lithium-ion-battery-capacity-to-grow-steadily-to-2030>. Rajat Verma, founder and chief executive officer of the Indian battery recycler Lohum, expects India's battery production to reach 70 percent LFP chemistries and 30 percent NCM chemistries. Estimates of greater LFP production are reinforced by more Indians likely preferring the lower cost of LFP cells. See Lee Allen, "India's Huge EV Battery Gigafactory Plans Could Spur Black Mass Imports and Cut Exports, Said Lohum Founder," Fastmarkets, February 12, 2024, <https://www.fastmarkets.com/insights/indias-ev-battery-gigafactory-plans-could-spur-black-mass-imports-and-cut-exports/>; Jyoti Gulia et al., "Lithium-Ion Battery (LiB) Manufacturing Landscape in India: Market Trends and Outlook," JMK Research & Analytics, Institute for Energy Economics and Financial Analysis, January 2022, 25, https://ieefa.org/sites/default/files/resources/Lithium-Ion-Battery-Manufacturing-Landscape-in-India_January-2022.pdf.

production potential for battery packs in two-wheeled and three-wheeled electric vehicles because downstream domestic demand incentivizes upstream domestic production.¹⁴ Lastly, like most countries, India lacks significant dedicated recycling capacity for electric vehicle batteries, but it does have a robust electronic waste recycling segment.

In short, while India's present role across the global battery supply chain is negligible, India could become a notable producer of certain goods in the global battery supply chain. These goods include the following:

1. Raw materials: bauxite, copper ore, natural graphite, synthetic graphite, iron ore, phosphate rock, fluorspar, and manganese ore
2. Precursor materials: aluminum, refined copper, iron sulfate, phosphoric acid, hydrofluoric acid, and manganese sulfate
3. Battery cells: LFP cells
4. Battery packs: packs for two-wheeled and three-wheeled vehicles
5. Recycled materials: black mass

Figure 6: Battery Supply Chain



The materials listed are not exhaustive, and the transformation of raw materials into precursor materials often involves the production of intermediate products. **Bolded materials** are materials for which India has existing production, while *italicized materials* are materials for which India has production potential.

Source: Author creation, based on the data sources included in Annex A.

¹⁴ Tom Moerenhout et al., "India's Potential in the Midstream of Battery Production," International Institute for Sustainable Development, September 2023, 6, <https://www.iisd.org/system/files/2023-09/india-potential-midstream-battery-production>.

Raw materials. India has high production potential in bauxite, copper ore, natural graphite, synthetic graphite, iron ore, fluorspar, phosphate rock, and manganese ore, all of which are raw materials necessary to produce different precursor materials. India already produces these raw materials, and it is a major global producer of bauxite, iron ore, and manganese ore (Table 1).¹⁵ Existing raw materials production is a major indicator of potential raw materials production as it means the country has the expertise and capital for producing certain raw materials. India also has reserves or economically recoverable mineral resources, of these raw materials, so it could continue to produce these raw materials.¹⁶

India has some resources of lithium, nickel, and cobalt too. While these raw materials are some of the highest-cost raw materials in electric vehicle batteries, they are not currently found in economically recoverable quantities in India and are thus not mined.¹⁷ These raw materials may not be economically recoverable for a variety of reasons, including the targeted element's price, mining capital expenditures, operating costs, metallurgical recovery rates, and deposit grade. However, mineral exploration has been lacking in India, and further exploration may qualify India's existing resources for these minerals into reserves, or it could discover additional mineral resources.¹⁸ Nonetheless, India is unlikely to find major reserves of lithium, nickel, and cobalt, meaning India has a low production potential in these raw materials.

pdf; PricewaterhouseCoopers Private Limited, "Giga-Scale Battery Manufacturing in India: Powering through Challenges in Domestic Production," Niti Aayog, September 2020, 7, <https://www.niti.gov.in/sites/default/files/2023-03/Giga-scale%20battery%20manufacturing%20in%20India%20Powering%20through%20challenges%20in%20domestic%20production%20Niti%20Aayog.pdf>; Gulia et al., "Lithium-Ion Battery (LiB) Manufacturing Landscape in India," 10; "India's EV Sales Surge to 1.5 Million in 2023, Up 50%: Report," *Economic Times*, January 19, 2024, <https://energy.economictimes.indiatimes.com/news/power/indias-ev-sales-surge-to-1-5-million-in-2023-up-50-report/106971357>; Ajit Dalvi, "Electric 3-Wheeler Sales Surge 57% in FY2024, Hit Record Levels of 630,000 Units," *Autocar Professional*, March 31, 2024, <https://www.autocarpro.in/analysis-sales/electric-3-wheeler-sales-surge-57-in-fy2024-hit-record-levels-of-630000-units--119850>; "New Scheme Announced to Push Sale of Electric Two-Wheelers and Three-Wheelers in India," *Economic Times*, March 13, 2024, <https://economictimes.indiatimes.com/industry/renewables/new-scheme-announced-to-push-sale-of-electric-two-wheelers-three-wheelers-in-india/articleshow/108464976.cms?from=mdr>. For a list of companies assembling battery packs, see Gulia et al., "Lithium-Ion Battery (LiB) Manufacturing Landscape in India," 11. Packs are usually different, such as by voltage and size, for different types of vehicles. Consequently, sourcing methods for two-wheeled and three-wheeled battery packs are different from four-wheeled battery packs. For example, two-wheeled and three-wheeled electric vehicle manufacturers often source packs from local manufacturers or have in-house assemblies, while four-wheeled electric vehicle manufacturers have in-house assemblies or import complete battery packs. See "Nexcharge Starts Mass Production of Li-Ion Battery Packs at Its New Plant in Gujarat," *Economic Times*, May 5, 2022, <https://auto.economictimes.indiatimes.com/news/auto-components/nexcharge-starts-mass-production-of-li-ion-battery-packs/91341880>; and Gulia et al., "Lithium-Ion Battery (LiB) Manufacturing Landscape in India," 23.

¹⁵ Ministry of Mines, "Annual Report 2022–23," 21, 250.

¹⁶ Ministry of Mines, "Annual Report 2022–23," 257–258.

¹⁷ International Energy Agency, "Global EV Outlook 2023," 61.

¹⁸ Rajesh Chadha, Ganesh Sivamani, and Karthik Bansal, "Assessing the Criticality of Minerals for India 2023," Centre for Social and Economic Progress, 2023, 18, <https://csep.org/wp-content/uploads/2023/04/Critical-Minerals-for-India-1-1.pdf>. In February 2023, the Geological Survey of India announced the discovery of a major lithium resource in Jammu and Kashmir. See Ministry of Mines, "Geological Survey of India Finds Lithium and Gold Deposits," Government of India, February 9, 2023, <https://pib.gov.in/PressReleaseDetailm.aspx?PRID=1897799>.

Table 1: India's Raw Materials Production and Proved Reserves

EXTRACTION			PROCESSING	CELL COMPONENT PRODUCTION
Raw material	Production in 2021-22 (P)	Proved reserves	Precursor material	Cell components
Bauxite	22,493,947 metric tons	560,865,000 metric tons	Aluminum (primary)	Positive current collector
Copper ore	3,569,632 metric tons	128,267,000 metric tons	Copper (refined)	Negative current collector
Natural graphite	57,264 metric tons	4,386,467 metric tons	Spherical graphite	Anode material
Synthetic graphite	>30,000 metric tons	N/A as synthetic graphite is produced from petroleum coke and coal tar.		
Fluorspar	1,237 metric tons	228,393 metric tons	Hydrofluoric acid	Electrolyte salt
Phosphate rock	1,395,079 metric tons	27,103,158 metric tons	Phosphoric acid	LFP cathode material
Iron ore	253,974,000 metric tons	4,631,786,000 metric tons	Iron sulfate	
Lithium	0	5,900,000 metric tons of resources	Lithium carbonate	
			Lithium hydroxide	NCM cathode material
Nickel	0	189,000,000 metric tons of resources	Nickel sulfate	
Cobalt	0	44,910,000 metric tons of resources	Cobalt sulfate	
Manganese	2,695,991 metric tons	61,510,000 metric tons	Manganese sulfate	

Table includes raw materials' corresponding precursor materials and cell components.

Source: Author creation based on data from the following sources: Ministry of Mines, "Annual Report 2022-23," Government of India, March 10, 2023, 250, 257-258, <https://mines.gov.in/admin/storage/app/uploads/6433da09a9f741681119753.pdf>; "Sourcing Local, Supplying Global," Epsilon Advanced Materials, accessed February 29, 2024, <https://www.epsilonam.com/>; Indian Bureau of Mines, "Graphite" in *Indian Minerals Yearbook 2021* (Nagpur: Indian Bureau of Mines, December 2022), https://ibm.gov.in/writereaddata/files/01312023164915Graphite_2021.pdf; Petroleum Planning & Analysis Cell, "Snapshot of India's Oil and Gas Data: Monthly Reckoner," Ministry of Petroleum and Natural Gas, November 2023, 6, 32, https://ppac.gov.in/uploads/whatsnew/1702638638_MRR_November%202023.pdf; Ministry of Mines, "Lithium Discovery in Jammu and Kashmir," July 31, 2023, <https://pib.gov.in/PressReleasePage.aspx?PRID=1944302>; Ministry of Petroleum and Natural Gas, "Annual Report 2022-23," Government of India, 2023, 32, <https://mopng.gov.in/files/TableManagements/annualreport2223.pdf>; Swansy Afonso, "First India EV Battery Plant Takes Shape to Cut China Dependence," *Bloomberg*, April 8, 2021, <https://www.bloomberg.com/news/articles/2021-04-08/first-india-ev-battery-plant-takes-shape-to-cut-china-dependence>.)

Precursor materials: India has high production potential in aluminum, refined copper, iron sulfate, phosphoric acid, hydrofluoric acid, and manganese sulfate. Like the raw materials, India has existing production of these precursor materials. While most

of India's production of precursor materials is not destined for the battery supply chain, existing production is again a major indicator of production potential as companies already have access to the capital, expertise, and raw materials to produce precursor materials. Importantly, India's precursor production for the battery supply chain may increase as India's demand grows for electric vehicle batteries and incentivizes domestic production of precursor materials.

For precursor materials that India does not currently produce, India has moderate production potential in spherical graphite, which is processed natural and synthetic graphite that eventually goes into anode material. India not only produces the raw materials necessary for producing spherical graphite, but it has also produced spherical graphite in trials.¹⁹ Regarding other precursor materials like lithium carbonate, India has low production potential because it lacks existing capacity and the corresponding raw materials. However, India's central government may be seeking to import the raw materials to produce precursor materials such as lithium carbonate. In 2019, it created the state-owned company Khanij Bidesh India Ltd (KABIL) to secure overseas mineral supplies, specifically minerals needed in batteries.²⁰ India could, just as China does, import these raw materials or intermediate products from overseas and process them domestically into precursor materials.

Cell components: India has moderate production potential in positive current collectors, negative current collectors, anode material, LFP cathode material, and electrolyte salt. India produces aluminum and refined copper, which is necessary to manufacture positive current collectors and negative current collectors, respectively. It also has some of the necessary precursor materials for other cell components, such as iron sulfate and phosphoric acid for LFP cathode material. Specifically, Himadri Specialty Chemicals intends to build a production facility for LFP cathode material in Odisha, and the government of India's International Advanced Research Centre for Powder Metallurgy and New Materials is collaborating with the Hyderabad-based company Altmin to produce LFP cathode material too.²¹ For anode material, the Indian company Epsilon Advanced Mate-

¹⁹ The company considering building a spherical graphite production facility in India is Tirupati. See Jacqueline Holman, "Insight Conversation: Shishir Poddar, Tirupati Graphite," S&P Global, March 4, 2022, <https://www.spglobal.com/commodityinsights/en/market-insights/blogs/metals/030422-ic-tirupati-graphite-shishir-poddar>; Tirupati Graphite, "Successful Trials of Battery-Grade Spherical Graphite," press release, February 22, 2021, https://www.rns-pdf.londonstockexchange.com/rns/7618P_1-2021-2-19.pdf; Tirupati Graphite, "Sustainability Report 2021," 2021, 5, <https://tirupatigraphite.co.uk/images/Sustainability%20Report%202021.pdf>.

²⁰ "KABIL (Khanij Bidesh India Limited)," Ministry of Mines, Government of India, accessed February 6, 2024, <https://mines.gov.in/webportal/content/kabil>. The Government of India is seeking to access lithium from Australian reserves and mines, and in January 2024, KABIL signed a \$24-million lithium exploration agreement in Argentina with a state-owned Argentinian company. See Abishek Law, "India Tapping into Australia for More Lithium Mines," *The Hindu BusinessLine*, December 1, 2023, <https://www.thehindubusinessline.com/economy/india-tapping-into-australia-for-more-lithium-mines/article67593334.ece>; Ministry of Mines, "India Signs Agreement for Lithium Exploration & Mining Project in Argentina," Government of India, January 15, 2024, <https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1996380>; "India's State-Owned KABIL Signs \$24 Mln Lithium Exploration Deal in Argentina," *Reuters*, January 15, 2024, <https://www.reuters.com/markets/commodities/indias-state-owned-kabil-signs-24-mln-lithium-exploration-deal-argentina-2024-01-15/>.

²¹ "India's First Commercial Lithium-Iron Phosphate Cathode Plant to Be Set up in Odisha," *Swarajya*, December 14, 2023, <https://swarajyamag.com/amp/story/business/indias-first-commercial-lithium-iron-phosphate-cathode-plant-to-be-set-up-in-odisha>; "Pilot Plant to Make Lithium-Ion Cell Raw Material to Open in Hyderabad," *The Hindu*, August 17, 2023, <https://www.thehindu.com/news/cities/Hyderabad/pilot-plant-to-make-lithium-ion-cell-raw-material-to-open-in-hyderabad/article67206549.ece>; "India's Push for Self-Reliance: ARCI Collaborates with Altmin to Produce Key Cathode Active Material," *The Hindu BusinessLine*, August 17, 2023, <https://www.thehindubusinessline.com/economy/india-forays-into-cathode-active>.

rial is building a production facility in Karnataka, and The Advanced Carbons Company, a subsidiary of HEG Limited, seeks to invest in a production facility for anode material.²² Gujarat Fluorochemicals and Neogen Chemicals intend to build electrolyte salt plants.²³

The cell component for which India has low production potential is NCM cathode material given India's lack of domestically produced lithium carbonate, nickel sulfate, and cobalt sulfate. India does mine manganese ore and produce manganese sulfate, which is necessary in NCM cathode material, and India could indeed import the lithium carbonate, nickel sulfate, and cobalt sulfate necessary to produce NCM cathode material.²⁴ KABIL is permitted to reach offtake (purchase) agreements with overseas mining companies. These offtake agreements could encompass KABIL purchasing precursor materials like lithium carbonate and transforming them into cathode material domestically.²⁵ Notably, India has a cost advantage in anode and cathode material production as India's infrastructure costs are up to 70 percent lower versus other top countries in chemical manufacturing.²⁶

Battery cells: India has high production potential in LFP battery cells, as evidenced by 70 percent of India's future battery production estimated to be LFP chemistries.²⁷ The Indian company Log9 Materials already produces LFP battery cells for two-wheeled and three-wheeled electric vehicles.²⁸ India also has moderate production potential

material-production-through-ppp/article67205957.ece. The Indian company Epsilon Advanced Materials has also acquired a technology center in Germany for producing LFP cathode material. See "EAM Completes Acquisition of LFP Cathode Technology Center in Germany, Plans for Commercial Scale Production for Global Supply Chain," *BusinessWire*, February 2, 2024, <https://www.businesswire.com/news/home/20240201488246/en/>.

²² Epsilon Advanced Materials, "Epsilon Advanced Materials (EAM) Announces Investment of \$650M Manufacturing Facility in North Carolina to Strengthen EV Battery Industry in the United States," October 26, 2023, https://www.epsilonam.com/pdf/EAM-NC-Announcement-press-release_Final.pdf; "Epsilon's Anode Manufacturing Plans and Recommendations to Support Component Manufacturing in India," *EVreporter*, October 11, 2023, <https://evreporter.com/epsilons-anode-manufacturing-plans-and-recommendation-to-support-component-manufacturing-in-india/>; Dhiyanesh Ravichandran, "HEG to Invest ₹ 900 Crores to Manufacture Graphite Anode for Li-Ion Batteries," *ETN News*, November 14, 2022, https://etn.news/index.php?option=com_easyblog&view=entry&id=2902&Itemid=562; Rishi Ranjan Kala, "TACC to Finalise Land for Anode Manufacturing Facility by December," *The Hindu BusinessLine*, December 7, 2022, <https://www.thehindubusinessline.com/companies/tacc-to-finalise-land-for-anode-manufacturing-facility-by-december/article66231019.ece>.

²³ "GFCL EV Products to Invest Rs 6,000 Cr in Next 4-5 Yrs to Ramp Up Production," *Economic Times*, February 7, 2024, <https://economictimes.indiatimes.com/industry/renewables/gfcl-ev-products-to-invest-rs-6000-cr-in-next-4-5-yrs-to-ramp-up-production/articleshow/107487647.cms?from=mdr>; "Neogen Ionics Completes Land Acquisition in Gujarat to Establish Battery Materials Facility at a Greenfield Site," *Economic Times*, December 19, 2023, <https://auto.economictimes.indiatimes.com/news/auto-components/neogen-ionics-completes-land-acquisition-in-gujarat-to-establish-battery-materials-facility-at-a-greenfield-site/106112509>.

²⁴ Dhruv Warrior, Akanksha Tyagi, and Rishabh Jain note, "India's mineral reserves are deficient in lithium, nickel, and cobalt, and importing these metals will be necessary if and when cathode active material production is set up in India later in the decade." See Dhruv Warrior, Akanksha Tyagi, and Rishabh Jain, "How Can India Indigenise Lithium-Ion Battery Manufacturing? Formulating Strategies across the Value Chain," Council on Energy, Environment and Water, February 2023, 13, <https://www.ceew.in/sites/default/files/how-can-india-scale-lithium-ion-battery-manufacturing-sector-and-supply-chain.pdf>.

²⁵ "Why Choose Us?" Khanij Bidesh India Limited (KABIL), accessed February 29, 2024, <https://kabilindia.in>.

²⁶ Avinash Goyal, "India: The Next Chemicals Manufacturing Hub," McKinsey & Company, March 2023, 10, <https://www.mckinsey.com/-/media/mckinsey/industries/chemicals/our%20insights/india%20the%20next%20chemicals%20manufacturing%20hub/india-the-next-chemicals-manufacturing-hub-vf.pdf?shouldIndex=false>.

²⁷ Allen, "India's Huge EV Battery Gigafactory Plans,"; Gulia et al., "Lithium-Ion Battery (LiB) Manufacturing Landscape in India," 25.

²⁸ Sameer Fayaz, "Log9 Materials Reveals First Made-in-India Battery Cell Line for EVs," *Hindustan Times*, April 21, 2023, <https://auto.hindustantimes.com/auto/electric-vehicles/log9-materials-reveals-first-made-in-india-battery-cell-for-electric-vehicles-41682068051681.html>; Sahil Kukreja, "Log9 Materials Launches India's First Lithium-Ion Cell Manufacturing Facility: Details," *Times of India*, April 24, 2023, <https://timesofindia.indiatimes.com/auto/policy-and-industry/log9-materials-launches-indias-first-lithium-ion-cell-manufacturing-facility-details/articleshow/99688828.cms>.

in NCM battery cells. Exide Industries, in collaboration with the Chinese company SVOLT Energy, is building a factory for both LFP and NCM battery cells in Bengaluru, which will supply two-wheeled and three-wheeled vehicles.²⁹ Ola Electric, which has the highest market share in India's electric two-wheeled vehicle segment, is building an NCM gigafactory with an initial capacity of 5 gigawatt hours (GWh) and a full-scale capacity of 100 GWh in Tamil Nadu.³⁰

India's high production potential in LFP battery cells and moderate production potential in NCM battery cells are bolstered by government subsidies. India's government offers battery manufacturing subsidies called the Production-Linked Incentive (PLI) Scheme for Advanced Chemistry Cell (ACC), which reduces companies' capital costs for battery cell factories and seeks to increase India's domestic battery cell manufacturing capacity by 50 GWh in five years.³¹ The PLI scheme's initial recipients for 30 GWh of battery capacity are Reliance New Energy Battery Storage (LFP battery cells), ACC Energy Storage (LFP battery cells), and Ola Cell Technologies (NCM battery cells), with initial production at these gigafactories slated for 2024.³²

State governments like Tamil Nadu are also offering subsidies to incentivize battery cell production.³³

Battery packs: India has high production potential in assembling battery packs for two-wheeled and three-wheeled electric vehicles given high downstream demand for

²⁹ Exide Industries, "Exide Industries – Investor Presentation," May 2023, 29–33, <https://docs.exideindustries.com/pdf/investor-presentation-may23.pdf>; Ishita Ayan Dutt, "Exide Industries to Start Lithium-Ion Cell Project's Phase-1 by FY25-End," *Business Standard*, August 8, 2023, https://www.business-standard.com/companies/news/exide-to-start-first-phase-of-lithium-ion-cell-project-by-end-of-next-fy-123080800691_1.html.

³⁰ Alisha Sachdev and Tanay Sukumar, "How Ola Electric Scored a 40% Market Share in December," *LiveMint*, January 3, 2024, <https://www.livemint.com/companies/news/how-ola-electric-scored-a-40-market-share-in-december-11704219620890.html>; "Ola Electric to Give India Its First Gigafactory Next Year: CEO Bhavish Aggarwal," *Economic Times*, August 16, 2023, <https://economictimes.indiatimes.com/industry/renewables/ola-electric-to-give-india-its-first-gigafactory-next-year-ceo-bhavish-aggarwal/articleshow/102774164.cms?from=mdr>. The factory will produce indigenously developed NCM battery cells. See Ola Electric, "Ola Unveils India's First Indigenously Made Lithium Ion-Cell," July 12, 2022, <https://blog.olaelectric.com/indias-first-indigenously-made-lithium-ion-cell/>; Chloe Cornish, "Ola Electric Heads for Two-Wheeler IPO in India's First EV Listing," *Financial Times*, January 17, 2024, <https://www.ft.com/content/59d54ef1-dc97-426e-9dfe-d296a717625a>. As another example, Tata Group in June 2023 announced it would build a 20 GWh battery manufacturing facility, capable of expanding by an additional 20 GWh. See Sumit Khanna, "India's Tata Group Signs \$1.6 Billion EV Battery Plant Deal," *Reuters*, June 2, 2023, <https://www.reuters.com/business/autos-transportation/indias-tata-group-signs-16-bln-ev-battery-plant-deal-2023-06-02/>.

³¹ "National Programme on Advanced Chemistry Cell (ACC) Battery Storage," Ministry of Heavy Industries, Government of India, accessed February 6, 2024, <https://pliac.in>; "National Programme on Advanced Chemistry Cell Battery Storage," Department of Heavy Industries, Government of India, accessed February 6, 2024, <https://evyatra.beeindia.gov.in/central-govt-initiative-details/departement-of-heavy-industries-dhi>.

³² Ministry of Heavy Industries, "Three Companies Signed Program Agreement under (PLI) Scheme for Advanced Chemistry Cell (ACC) Battery Storage," Government of India, July 29, 2022, <https://pib.gov.in/PressReleasePage.aspx?PRID=1846078>; Shraddha Kakade, "What's Next for India's Battery Manufacturing Industry in 2024?" *ETN News*, January 16, 2024, <https://etn.news/energy-storage/what-s-next-for-india-s-battery-manufacturing-industry-in-2024>; Saurav Anand, "Commercial Production of ACC Batteries under PLI Scheme to Commence from FY24-End," *LiveMint*, August 24, 2023, <https://www.livemint.com/auto-news/commercial-production-of-acc-batteries-under-pbi-scheme-to-commence-from-fy24end-11692890717885.html>. Ola Cell Technologies is building its facility in Krishnagiri, Tamil Nadu. ACC Energy Storage is building its facility in Dharwad, Karnataka. Reliance New Energy Battery Storage is building its facility in Jamnagar, Gujarat.

³³ Industries, Investment Promotion and Commerce Department, "Tamil Nadu: Electric Vehicles Policy," Government of Tamil Nadu, 2023, https://investingintamilnadu.com/DIGIGOV/StaticAttachment?AttachmentFileName=/pdf/poli_noti/TN_Electric_Vehicles_Policy_2023.pdf; Sanjay Vijayakumar, "Can Tamil Nadu Accelerate into the Future with Electric Vehicles?" *The Hindu*, October 29, 2023, <https://www.thehindu.com/news/national/tamil-nadu/can-tamil-nadu-accelerate-into-the-future-with-electric-vehicles/article67471289.ece>.

these vehicles and government subsidies for purchasing these vehicles, which incentivizes domestic battery production for two-wheeled and three-wheeled electric vehicles. Indian companies already manufacture these battery packs, but mainly with cells imported from China.³⁴ Companies converting battery cells to battery packs in India include Exide Industries, Amara Raja, Aether, and Ola Electric.³⁵ For example, Nexcharge, a joint venture between Exide Industries and the Swiss company Leclanché SA, has a battery pack facility in Gujarat.³⁶ Smaller manufacturers in India also produce battery packs.³⁷ The capital costs for battery assembly are low, with the minimum investment only being \$1.3 million in some cases.³⁸ The expertise required to manufacture battery packs is also relatively low compared to other segments of the battery supply chain.³⁹ Despite these low barriers to entry, India's share of global battery pack assembly is low.⁴⁰

However, India's high production potential in battery packs for two-wheeled and three-wheeled vehicles is supported by the Indian central government's Electric Mobility Promotion Scheme 2024, which offers purchase subsidies for two-wheeled and three-wheeled electric vehicles with traction battery packs assembled in India.⁴¹ State governments like Karnataka are also offering incentives for battery pack manufacturing, such as exemptions from stamp duties and offers of interest-free loans.⁴² Increasing Indian demand for two-wheeled and three-wheeled vehicles will further support increased domestic production of battery packs for two-wheeled and three-wheeled electric vehicles. Two-wheeled and three-wheeled electric vehicles have higher existing and projected market penetration than four-wheeled electric vehicles in India, with

³⁴ Moerenhout et al., "India's Potential in the Midstream of Battery Production," 6; PricewaterhouseCoopers Private Limited, "Giga-Scale Battery Manufacturing in India," 7-12; Gulia et al., "Lithium-Ion Battery (LiB) Manufacturing Landscape in India," 5, 10; Srinath Rengarajan, "Battery Manufacturing In India: Time for a Bharatvolt?" *EV Tech News*, April 2, 2021, <https://evtechnews.in/battery-manufacturing-in-india-time-for-a-bharatvolt/>; Shivam Chauhan et al., "Knowledge Paper on Lithium-Ion Battery Assembling in India," India Energy Storage Alliance, accessed February 29, 2024, 5, https://indiaesa.info/media/downloadfiles/whitepaper_final_lowres._993448987.pdf. For a list of companies assembling battery packs, see Gulia et al., "Lithium-Ion Battery (LiB) Manufacturing Landscape in India," 11.

³⁵ Gulia et al., "Lithium-Ion Battery (LiB) Manufacturing Landscape in India," 12, 22-23.

³⁶ Dhyanesh Ravichandran, "Nexcharge Begins Mass Production of Li-Ion Battery at Its Gujarat Facility," *ETN News*, May 5, 2022, <https://etn.news/buzz/nexcharge-lithium-ion-battery-production-india-prantij-facility>; Prathik Desai, "Nexcharge Starts Making Lithium-Ion Batteries in Gujarat," *Deccan Herald*, May 5, 2022, <https://www.deccanherald.com/business/nexcharge-starts-making-lithium-ion-batteries-in-gujarat-1106771.html>.

³⁷ Gulia et al., "Lithium-Ion Battery (LiB) Manufacturing Landscape in India," 23.

³⁸ Gulia et al., "Lithium-Ion Battery (LiB) Manufacturing Landscape in India," 18.

³⁹ Moerenhout et al., "India's Potential in the Midstream of Battery Production," 6.

⁴⁰ PricewaterhouseCoopers Private Limited, "Giga-Scale Battery Manufacturing in India," 10.

⁴¹ "Govt Brings New Rs 500 Crore EV Subsidy Scheme from April 1 after FAME. Check Details," *Economic Times*, April 2, 2024, <https://economictimes.indiatimes.com/industry/renewables/govt-brings-new-rs-500-crore-ev-subsidy-scheme-from-april-1-after-fame-check-details/articleshow/108960224.cms?from=mdr>; Twesh Mishra, "Localisation, Warranty Norms Stricter in Electric Mobility Promotion Scheme," *Economic Times*, March 15, 2024, <https://economictimes.indiatimes.com/industry/renewables/localisation-warranty-norms-stricter-in-electric-mobility-promotion-scheme/articleshow/108503643.cms?from=mdr>. Previously, India's Faster Adoption and Manufacturing of (Hybrid &) Electric Vehicles (FAME II) offered purchase subsidies for electric vehicles with traction battery packs assembled in India. See Ministry of Heavy Industries, "FAME India: Faster Adoption and Manufacturing of (Hybrid &) Electric Vehicles in India," July 11, 2022, 2, <https://static.pib.gov.in/WriteReadData/specifidocs/documents/2022/jul/doc202271169601.pdf>; Ministry of Heavy Industries, "Fame India Scheme," July 25, 2022, <https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1942506>; Gulia et al., "Lithium-Ion Battery (LiB) Manufacturing Landscape in India," 15; Ministry of Heavy Industries, "Phased Manufacturing Program (PMP) for xEV Parts for Eligibility under FAME India Scheme Phase-II – Revision – Regarding," Government of India, January 10, 2021, https://fame2.heavyindustries.gov.in/WriteReadData/Pdf/Revised_PMP%201_10_2021.pdf.

⁴² Gulia et al., "Lithium-Ion Battery (LiB) Manufacturing Landscape in India," 32.

two-wheeled and three-wheeled vehicles representing between 90 percent and 95 percent of total electric vehicle sales in India in 2023.⁴³

Battery recycling: India currently has companies with moderate production potential in battery recycling. These companies have existing recycling capacity for electronic scrap and waste but also electric vehicle batteries. Indian companies like Attero have announced battery recycling projects, and India has established regulations, namely the Battery Waste Management Rules of 2022, to govern the recycling of these batteries.⁴⁴ Indian industry, however, lacks the processing capacity to refine black mass into precursor materials at commercial volumes.⁴⁵ As Indian society adopts more electric vehicles, more batteries will reach the end of their useful lives, creating more supply of recycled materials. Notably, India's volume of end-of-life batteries may grow more quickly than other countries given the shorter lifetimes of batteries for two-wheeled and three-wheeled vehicles.⁴⁶

4. CHALLENGES CONFRONTING INDIA'S BATTERY SUPPLY CHAINS

Despite its production potential in several segments of the battery supply chain, India faces three main challenges in increasing its production in these supply chain segments. First, India lacks reserves of nickel, cobalt, and lithium, which are high-cost raw materials in electric vehicle batteries.⁴⁷ All battery chemistries require lithium, and NCM batteries require nickel and cobalt. Given the high cost of these raw materials, com-

⁴³ "EV Penetration to Reach Up to 8% in Two-Wheelers and 16% in Three-Wheelers in FY2025 Says ICRA," *Financial Express*, March 15, 2024, <https://www.financialexpress.com/business/express-mobility-ev-penetration-to-reach-up-to-8-in-two-wheelers-and-16-in-three-wheelers-in-fy2025-says-icra-3426362/#>; Shubhangi Bhatia, "EV Penetration to Grow Exponentially, Sector's Profitability Expected to Delay Take off," *Economic Times*, November 7, 2023, <https://auto.economictimes.indiatimes.com/news/industry/ev-penetration-to-grow-exponentially-sectors-profitability-expected-to-delay-take-off/105009256>; Mohan Gupta, "Segmented Analysis of Electric Vehicle Penetration in India: 2-Wheelers, 3-Wheelers, and 4-Wheelers Compared," *EMobility+*, October 26, 2023, <https://emobilityplus.com/2023/10/26/segmented-analysis-of-electric-vehicle-penetration-in-india-2-wheelers-3-wheelers-and-4-wheelers-compared/>; Ajit Dalvi, "Electric 3-Wheeler Sales Surge 57% in FY2024, Hit Record Levels of 630,000 Units," *Autocar Professional*, March 31, 2024, <https://www.autocarpro.in/analysis-sales/electric-3-wheeler-sales-surge-57-in-fy2024-hit-record-levels-of-630000-units--119850>; Sibi Arasu, "India Is One of the World's Fastest-Growing EV Markets. This Is Why," *AP*, August 31, 2023, <https://apnews.com/article/electric-vehicles-india-boom-rickshaws-55e7ed315fc41d0ba56461386ad44a4b>; "India's EV Sales Surge to 1.5 Million in 2023, Up 50%: Report," *Economic Times*, January 19, 2024, <https://energy.economictimes.indiatimes.com/news/power/indias-ev-sales-surge-to-1-5-million-in-2023-up-50-report/106971357>. Displaying the higher relative demand for two-wheeled and three-wheeled electric vehicles over four-wheeled electric vehicles, FAME II provided subsidies to more two-wheeled (1.2 million) and three-wheeled electric vehicles (141,000) than four-wheeled electric vehicles (16,991). See "New Scheme Announced to Push sale of Electric Two-Wheelers and Three-Wheelers in India," *Economic Times*, March 13, 2024, <https://economictimes.indiatimes.com/industry/renewables/new-scheme-announced-to-push-sale-of-electric-two-wheelers-three-wheelers-in-india/articleshow/108464976.cms?from=mdr>.

⁴⁴ Lee Allen, "Recycled Metals to Help Feed India's Lithium Battery Boom by 2026: MRAI Conference," *Fastmarkets*, February 7, 2023, <https://www.fastmarkets.com/insights/recycled-metals-to-feed-indias-lithium-battery-boom/>; Kushal Dev Sharma, "How One Company Is Transforming Battery Recycling Business in India," *Economic Times*, August 6, 2023, <https://economictimes.indiatimes.com/news/india/how-one-company-is-transforming-battery-recycling-business-in-india/articleshow/102411868.cms?from=mdr>; "Omega Seiki, Attero Partner to Recycle over 100 MWh Batteries by 2028: Details," *Times of India*, February 6, 2024, <https://timesofindia.indiatimes.com/auto/news/omega-seiki-attero-partner-to-recycle-over-100-mwh-batteries-by-2028/articleshow/107458427.cms?from=mdr#>; Central Pollution Control Board, "Ministry of Environment, Forest and Climate Change Notification," Government of India, August 22, 2022, <https://cpcb.nic.in/uploads/hwmd/Battery-WasteManagementRules-2022.pdf>.

⁴⁵ Allen, "Recycled Metals to Help Feed India's Lithium Battery Boom by 2026"; Bhupesh Verma and Anjali Singh, "Recycling of EV Batteries in India: What Opportunity Lies Ahead?" *Mobility Outlook*, October 2021, <https://www.mobilityoutlook.com/commentary/recycling-of-ev-batteries-in-india-what-opportunity-lies-ahead/>.

⁴⁶ Sharma, "How One Company Is Transforming."

⁴⁷ International Energy Agency, "Global EV Outlook 2023," 61.

panies producing NCM batteries sometimes seek to vertically integrate into upstream mineral production to better control costs, such as by acquiring ownership stakes in raw materials projects.⁴⁸ However, India's lack of domestic reserves for these raw materials restricts vertical integration.⁴⁹ Extensive exploration, which could take years, may be required before India's resources can be qualified as reserves; even then, more years and capital will be required to convert the reserves into mined ore.

A major bottleneck in the battery supply chain is sourcing sufficient raw materials.⁵⁰ So far, India's lack of production in some minerals has not adversely impacted its battery supply chain given limited battery production and sufficient access to overseas raw materials.⁵¹ Yet, India's access to sufficient raw materials may become a challenge as Indian battery production increases, other companies, especially Chinese companies, secure overseas mineral production, and geopolitical tensions intensify.⁵² If India is unable to secure sufficient raw materials domestically or overseas, the lack of these raw materials — especially lithium — may undermine production in downstream segments of the battery supply chain, such as cathode manufacturing.⁵³

Second, production facilities across the battery supply chain from mines to gigafactories are capital-intensive, requiring hundreds of millions to billions of dollars in upfront capital investment. To illustrate the cost of raw material production facilities, China's Zhejiang Huayou Cobalt bought a hard rock lithium mine in Zimbabwe for \$422 million and built an accompanying \$300 million concentrating facility that can process 4.5 million metric tons of lithium ore into lithium concentrate annually.⁵⁴ Similarly, for precursor materials, Chilean company Sociedad Química y Minera (SQM) invested \$140 million to build a lithium processing plant in China that can produce 30,000 metric tons of lithium hydroxide annually.⁵⁵ Production facilities for cathode material and anode material are also costly. The Chinese electric vehicle company BYD announced its plan to build a \$290 million cathode facility in Chile that can produce 50,000 metric tons of LFP cathode annually, and the Indian company Epsilon Advanced Materials intends to build a \$650 million anode facility in the United States that can produce over 45,000 metric tons of anode material annually.⁵⁶ Battery cell plants are generally the most expensive: Panasonic's battery cell plant in Kansas will

⁴⁸ Charles Chang et al., "China's Global Reach Grows behind Critical Minerals," S&P Global, August 2023, 2, 7, 9, https://www.spglobal.com/_division_assets/images/special-editorial/china-s-global-reach-grows-behind-critical-minerals/082423-china-s-global-reach-grows-behind-critical-minerals.pdf.

⁴⁹ PricewaterhouseCoopers Private Limited, "Giga-Scale Battery Manufacturing in India," 5.

⁵⁰ PricewaterhouseCoopers Private Limited, "Giga-Scale Battery Manufacturing in India," 24.

⁵¹ Moerenhout et al., "India's Potential in the Midstream of Battery Production," 14.

⁵² Moerenhout et al., "India's Potential in the Midstream of Battery Production," 14; Pramoda Gode, Georg Bieker, and Anup Bandivadekar, "Battery Capacity Needed to Power Electric Vehicles in India from 2020 to 2035," International Council on Clean Transportation, February 2021, 10, <https://theicct.org/wp-content/uploads/2021/06/Battery-capacity-ev-india-feb2021.pdf>.

⁵³ Moerenhout et al., "India's Potential in the Midstream of Battery Production," 13.

⁵⁴ Farai Mutsaka, "A Chinese Mining Company Has Opened a Giant Lithium Processing Plant in Zimbabwe," *AP*, July 5, 2023, <https://apnews.com/article/lithium-processing-plant-chinese-92ccd10bd4ba028bc38fdadcdff06da>.

⁵⁵ Sociedad Química y Minera, "Corporate Presentation," August 2023, 11, https://s25.q4cdn.com/757756353/files/doc_presentations/2023/August-2023_Corporate-Presentation.pdf.

⁵⁶ "China EV Maker BYD to Build \$290 Million Battery Component Plant in Chile," *Reuters*, April 20, 2023, <https://www.reuters.com/business/autos-transportation/china-ev-maker-byd-build-290-mln-battery-component-plant-chile-2023-04-21/>; Paul Lienert, "India's EAM to Build \$650-Million EV Battery Materials Plant in US," *Reuters*, October 26, 2023, <https://www.reuters.com/business/autos-transportation/indias-eam-build-650-million-ev-battery-materials-plant-us-2023-10-26/>.

cost an estimated \$4 billion and produce an estimated 30 GWh of battery cells annually.⁵⁷ Battery pack assembly lines vary in cost depending on their capacity, ranging, for example, from \$1.3 million to \$160 million.⁵⁸

Adding to the capital challenges, upstream production facilities, like mines and processing facilities, can take several years to become revenue-producing assets. The global average timeline from deposit discovery to commercial mine production is 16 years.⁵⁹ Mineral production then takes months or even years to ramp up to nameplate capacity. For instance, Western companies seeking to use an advanced processing method for nickel and cobalt took at least five years on average to ramp up to capacity.⁶⁰ Many companies lack the capital to wait such a long time for their assets to generate revenue. Therefore, many companies prefer to buy battery cells and packs from overseas suppliers, instead of producing batteries themselves.⁶¹

Third, every supply chain step in the EV battery chain, from extraction to battery pack production, requires specialized expertise. Foreign companies and governments, namely China, have been investing in and developing production capabilities in the battery supply chain for decades. For instance, the Chinese company BYD, currently the world's largest electric vehicle producer and second largest producer of electric vehicle batteries, was founded in 1995, and the Chinese government began prioritizing research and development in electric vehicle technology in its Five-Year Plan in 2001.⁶² In 2009, the Chinese government then started significantly subsidizing the industry, providing an estimated \$29 billion in subsidies from 2009 to 2022.⁶³ Therefore, developing expertise along the battery supply chain takes substantial capital and time.

India's companies and government will face challenges in quickly developing indigenous production in the battery supply chain.⁶⁴ For example, Tata Motors sources its electric vehicle batteries from Tata AutoComp Systems, which is a Tata joint venture

⁵⁷ Andrew J. Hawkins, "Panasonic Breaks Ground on \$4 Billion EV Battery Plant in Kansas," *The Verge*, November 2, 2022, <https://www.theverge.com/2022/11/2/23437072/panasonic-kansas-ev-battery-factory-tesla-2170>; "Panasonic Energy's New EV Battery Facility in Kansas Will Initially Produce 30 GWh," Panasonic, October 31, 2022, <https://na.panasonic.com/us/news/panasonic-energys-new-ev-battery-facility-kansas-will-initially-produce-30-gwh>.

⁵⁸ Gulia et al., "Lithium-Ion Battery (LiB) Manufacturing Landscape in India," 18; Kalea Hall, "GM Plans to Invest Millions at Orion for New Battery Pack Line," *Detroit News*, December 8, 2021, <https://www.detroitnews.com/story/business/autos/general-motors/2021/12/08/gm-invest-millions-orion-new-battery-pack-line/6435282001/>. PricewaterhouseCoopers Private Limited, "Giga-Scale Battery Manufacturing in India," 24.

⁵⁹ Paul Manalo, "Discovery to Production Averages 15.7 Years for 127 Mines," S&P Global, June 6, 2023, <https://www.spglobal.com/marketintelligence/en/news-insights/research/discovery-to-production-averages-15-7-years-for-127-mines>.

⁶⁰ Angela Durrant, "The Rise and Rise of Indonesian HPAL – Can It Continue?" Wood Mackenzie, April 4, 2023, <https://www.woodmac.com/news/opinion/rise-of-indonesian-hpal/>. For instance, battery manufacturing facilities may only average 60 percent of nameplate capacity in the year after commissioning. See Andreas Breiter, "Power Spike: How Battery Makers Can Respond to Surging Demand from EVs," McKinsey & Company, October 18, 2022, 3, https://www.mckinsey.com/capabilities/operations/our-insights/power-spike-how-battery-makers-can-respond-to-surging-demand-from-evs#.

⁶¹ Breiter et al., "Power Spike," 7.

⁶² Arjun Kharpal and Evelyn Cheng, "Musk Once Laughed off BYD as a Threat. Now the Chinese Giant Has Taken Tesla's EV Crown — Here's How," *CNBC*, January 4, 2024, <https://www.cnbc.com/2024/01/05/how-byd-grew-from-a-phone-battery-maker-to-ev-giant-taking-on-tesla.html>; SNE Research, "From Jan to Sep in 2023"; "About BYD," BYD, accessed February 6, 2024, https://www.bydglobal.com/cn/en/BYD_ENAboutByd/CompanyIntro_mob.html; Wang Tao, "Recharging China's Electric Vehicle Policy," Carnegie-Tsinghua Center for Global Policy, August 2013, 2, https://carnegieendowment.org/files/china_electric_vehicles.pdf; Zeyi Yang, "How Did China Come to Dominate the World of Electric Cars?" *MIT Technology Review*, February 21, 2023, <https://www.technologyreview.com/2023/02/21/1068880/how-did-china-dominate-electric-cars-policy/>.

⁶³ Yang, "How Did China Come to Dominate the World of Electric Cars?"

⁶⁴ PricewaterhouseCoopers Private Limited, "Giga-Scale Battery Manufacturing in India," 24.

with the Chinese company Guoxuan High-Tech.⁶⁵ Major American automakers similarly rely on non-American companies for batteries and battery manufacturing technology.⁶⁶ Also, demand is soaring for skilled labor in India's electric vehicle industry, but India lacks sufficient labor with the necessary technical expertise.⁶⁷ Skilled labor shortages can cause commissioning and ramp-up delays, and it is affecting battery manufacturers globally, even incumbent battery companies.⁶⁸

5. OPPORTUNITIES FOR INTERNATIONAL COLLABORATION

To reduce dependence on China in the global battery supply chain, opportunities for international collaboration exist between India, other countries, and international financial institutions. First, India could export raw materials, precursor materials, LFP battery cells, battery packs for two-wheeled and three-wheeled vehicles, and black mass to other countries.⁶⁹ For example, overseas producers of NCM cathode material may lack sufficient manganese sulfate in their home countries or seek to diversify their suppliers; in such a case, India could export manganese sulfate to these overseas producers. Similarly, India, which already exports lead-acid batteries to many countries, could export LFP battery cells and battery packs for two-wheeled and three-wheeled vehicles to countries in South Asia and Southeast Asia. It could continue to export black mass for recycling to South Korea too.⁷⁰

Second, India could import end-of-life batteries from other countries and process them into black mass domestically for domestic use or export them to other markets. Many governments are mandating recycling for end-of-life batteries, but these countries lack adequate recycling capacity. India could import these end-of-life batteries, converting them into black mass that can be processed domestically or, as already occurs, exported overseas.⁷¹ India currently lacks commercial production in converting black mass into precursor materials, like lithium carbonate, but with such production, Indian recycling companies could supply recycled battery materials to companies seeking or mandated to use recycled materials (not virgin materials) in their electric vehicle batteries.⁷² Im-

⁶⁵ Aditi Shah, "Tata Motors Says 20% Rise in Battery Cell Costs Increasing Short-Term Pressure," *Reuters*, March 29, 2022, <https://www.reuters.com/business/autos-transportation/tata-motors-says-20-rise-battery-cell-costs-increasing-short-term-pressure-2022-03-29/>.

⁶⁶ Michael Wayland, "Ford to Scale Back Plans for \$3.5 Billion Michigan Battery Plant As EV Demand Disappoints, Labor Costs Rise," *CNBC*, November 21, 2023, <https://www.cnn.com/2023/11/21/ford-scales-back-ev-battery-plant-in-michigan.html>; Tom Krisher, "GM, Samsung Plan New EV Battery Cell factory in US," *AP*, April 25, 2023, <https://apnews.com/article/gm-samsung-korea-battery-c9a0c619100fe102ffe3c074ae33f746>.

⁶⁷ PricewaterhouseCoopers Private Limited, "Giga-Scale Battery Manufacturing in India,"; Prachi Verma, "Surge in Demand for Skilled Electric Vehicle Talent As Indian Manufacturers Race to Meet Green Mobility Demands," *Economic Times*, January 14, 2024, <https://economictimes.indiatimes.com/industry/renewables/surge-in-demand-for-skilled-electric-vehicle-talent-as-indian-manufacturers-race-to-meet-green-mobility-demands/articleshow/106819844.cms>.

⁶⁸ Jakob Fleischmann et al., "Battery 2030: Resilient, Sustainable, and Circular," McKinsey & Company, 2023, 6, <https://www.mckinsey.com/-/media/mckinsey/industries/automotive%20and%20assembly/our%20insights/battery%202030%20resilient%20sustainable%20and%20circular/battery-2030-resilient-sustainable-and-circular.pdf?shouldIndex=false>; Breiter et al., "Power Spike," 3–5; Nikhil Bhandari et al., "Batteries: The Greenflation Challenge," Goldman Sachs, March 8, 2022, 25, <https://www.goldmansachs.com/intelligence/pages/gs-research/batteries-the-greenflation-challenge/report.pdf>.

⁶⁹ Already, the American electric vehicle company Tesla imports over \$1 billion in components from India annually. See "Tesla to Double Its Components Imports from India - Trade Minister," *Reuters*, November 13, 2023, <https://www.reuters.com/business/autos-transportation/tesla-about-double-its-components-imports-india-trade-minister-2023-11-14/>.

⁷⁰ Allen, "Recycled Metals to Help Feed India's Lithium Battery Boom by 2026,"; Verma and Singh, "Recycling of EV Batteries In India."

⁷¹ Allen, "Recycled Metals to Help Feed India's Lithium Battery Boom by 2026."

⁷² Allen, "Recycled Metals to Help Feed India's Lithium Battery Boom by 2026."

portantly, India should establish standards on importing end-of-life batteries, especially regarding their safe storage, and it should also impose tariffs on imported end-of-life batteries to incentivize the recycling of end-of-life batteries already present in India.

Third, countries seeking investment in their minerals sector could seek investment from India's central government. As previously noted, India is seeking to invest in overseas mineral resources through KABIL, and it is heavily targeting cobalt and lithium.⁷³ Currently, it is exploring lithium resources in Argentina, ultimately seeking to develop them into mines, and it is also reportedly in discussions to acquire lithium assets in Bolivia.⁷⁴ India could also leverage its membership in the Minerals Security Partnership to vet overseas mineral projects for investment or partner with other member governments in investing in overseas mineral projects. The best candidate countries for Indian investment are those countries with mineral resources but lacking domestic investment or seeking non-Chinese investment, such as Australia and countries in Africa.

Fourth, international financial institutions and other governments' development financial institutions could financially support battery supply chain projects in India.⁷⁵ Many of these financial institutions are targeting projects that support the electric vehicle transition, and they could offer such projects loans and other financial products, including loan guarantees, political risk insurance, supporting infrastructure investments, and technical assistance like grants for feasibility studies and environmental impact studies. If a project receives support from international financial institutions, other financial actors, such as the private sector, will likely be more willing to financially support the project as they view the project as partially de-risked. One specific foreign government entity that may be interested in investing in Indian projects for raw materials and precursor materials is the Japan Organization for Metals and Energy Security (JOGMEC), which has invested in several mineral-related projects overseas.⁷⁶ Further down the supply chain, the Japan Bank for International Cooperation (JBIC) and the Japan External Trade Organization (JETRO) could be financing and investment partners for Indian projects, such as cathode material production. Crucially, JBIC is already financing an electric vehicle project in Uttar Pradesh.⁷⁷

Fifth, India and other countries could increase their university-to-university and govern-

⁷³ Ministry of Mines, "Annual Report 2022–23," 49.

⁷⁴ Ministry of Mines, "India Signs Agreement for Lithium Exploration & Mining Project in Argentina," "India's State-Owned KABIL Signs \$24 Mln Lithium Exploration Deal in Argentina," "India in Talks with Argentina, Bolivia to Acquire Lithium Assets – Secretary," *Reuters*, December 19, 2023, <https://www.reuters.com/markets/commodities/india-talks-with-argentina-bolivia-acquire-lithium-assets-secretary-2023-12-19/>.

⁷⁵ For instance, the World Bank's Climate-Smart Mining Initiative launched in 2019, and it seeks to support "the sustainable extraction and processing of minerals and metals to secure supply for clean energy technologies." See "Climate-Smart Mining: Minerals for Climate Action," World Bank, accessed February 8, 2024, <https://www.worldbank.org/en/topic/extractiveindustries/brief/climate-smart-mining-minerals-for-climate-action>.

⁷⁶ Mamoru Tsuge and Shoya Okinaga, "Japan Takes First Stake in Heavy Rare Earths to Reduce China Reliance," *Nikkei Asia*, March 9, 2023, <https://asia.nikkei.com/Spotlight/Supply-Chain/Japan-takes-first-stake-in-heavy-rare-earths-to-reduce-China-reliance>; "Japan's JOGMEC, Sojitz Invest \$9 Million in Rare-Earths Miner Lynas," *Reuters*, September 20, 2022, <https://www.reuters.com/markets/commodities/japans-jogmec-sojitz-invest-9-mln-rare-earths-miner-lynas-2022-09-20/>. Notably, the Japanese entities saved the Australian rare earth company Lynas from collapse in 2016 through debt relief and additional funds. See "Lenders Rescue Rare Earths Producer Lynas from Collapse," *Reuters*, October 25, 2016, <https://www.reuters.com/article/lynas-corp-debtrenegotiation-idUSL4N1CV674>.

⁷⁷ "Electric Vehicle Project," reference number 2022-0156, Japan Bank for International Cooperation, March 13, 2023, <https://www.jbic.go.jp/en/business-areas/environment/projects/63850.html>.

ment-to-government research collaboration. With university collaboration, for example, the Indian Institutes of Technology (IITs) could collaborate with Japan's University of Tokyo and the United States' Massachusetts Institute of Technology given their existing battery research.⁷⁸ Regarding government collaboration, India's Ministry of Mines and Australia's Commonwealth Scientific and Industrial Research Organisation (CSIRO) could expand their existing research collaboration on critical minerals, and India's government could collaborate with other governments' research entities, like the U.S. government's national laboratories.⁷⁹ The resulting innovation from this research collaboration could be made available to commercial entities in India and the collaborating countries.

6. CONCLUSION

As the world transitions to electric vehicles, India could advance this transition by playing a producer role in certain segments of the global battery supply chain. Currently, India lacks a sizable production share in any segment of the global battery supply chain, which is dominated by China. India, however, has existing production and significant growth potential in specific segments of the battery supply chain, especially as some countries seek to diversify their supply chains away from China due to geopolitical concerns. For instance, India already produces several of the ancillary raw and precursor materials needed for battery manufacturing but uses them for other goods. Examples of these raw materials include bauxite, copper ore, and iron ore, while examples of these precursor materials include aluminum, refined copper, and iron sulfate.

Due to India's lack of lithium, nickel, and cobalt reserves, India is unlikely to produce these raw materials, their corresponding precursor materials, and NCM cathode material. However, India could discover more mineral resources, qualify existing mineral resources into reserves, import these materials, use investment and offtake agreements to secure such materials overseas, or increase the recovery of these materials through recycling. In addition to increasing international trade of battery materials and developing domestic technical expertise, capital will be crucial for developing production capacity in India due to the capital intensity of battery supply chain projects. Together, international trade, investment, research collaboration, and finance from development finance institutions could help incorporate India and its companies into the global supply chain for electric vehicle batteries.

⁷⁸ Debika Chakraborty, "IIT Madras Researchers Develop New Battery Technology for Electric Cars," *NDTV*, May 30, 2022, <https://www.ndtv.com/education/iit-madras-researchers-develop-new-battery-technology-for-electric-cars-3023022>; Jota Tsuchiya, "Japanese Scientists Make Better Lithium-Ion Battery without Cobalt," *Nikkei Asia*, October 29, 2023, <https://asia.nikkei.com/Business/Science/Japanese-scientists-make-better-lithium-ion-battery-without-cobalt>; Ann Trafton, "Cobalt-Free Batteries Could Power Cars of the Future," *MIT News*, January 18, 2024, <https://news.mit.edu/2024/cobalt-free-batteries-could-power-future-cars-0118>.

⁷⁹ "India-CSIRO Partnerships as Part of the India Economic Strategy (IES)," Commonwealth Scientific and Industrial Research Organisation, Australian Government, accessed February 8, 2024, <https://www.csiro.au/en/work-with-us/international/asia/south-asia/india>; "CSIRO's Science Underpinning the Australia-India Critical Minerals Partnership," Commonwealth Scientific and Industrial Research Organisation, Australian Government, accessed February 8, 2024, <https://www.csiro.au/en/work-with-us/industries/mining-resources/resourceful-magazine/issue-27/australia-indian-partnership>. The United States and India have "pledged to hasten bilateral collaboration to secure resilient critical minerals supply chains through enhanced technical assistance and greater commercial cooperation, and exploration of additional joint frameworks as necessary." See White House, "Joint Statement from the United States and India," June 22, 2023, <https://www.whitehouse.gov/briefing-room/statements-releases/2023/06/22/joint-statement-from-the-united-states-and-india/>.

ANNEX A

Many sources contributed to the creation of Figure 6 on the battery supply chain and analysis of India's production capacity and potential.

For latest figures on India's production of aluminum, bauxite, copper ore, fluor spar, graphite, iron ore, manganese, manganese sulfate, natural graphite refined copper, phosphate rock, and India's potential production of cobalt, nickel, see Ministry of Mines, "Annual Report 2022–23," Government of India, March 10, 2023, <https://mines.gov.in/admin/storage/app/uploads/6433da09a9f741681119753.pdf>.

For figures on India's production of natural and synthetic graphite, see Fastmarkets, accessed March 6, 2024, <https://www.fastmarkets.com/industrial-minerals/graphite>; Indian Bureau of Mines, Indian Minerals Yearbook 2021 (Nagpur: Indian Bureau of Mines, December 2022), https://ibm.gov.in/writereaddata/files/01312023164915Graphite_2021.pdf; Swansy Afonso, "First India EV Battery Plant Takes Shape to Cut China Dependence," Bloomberg, April 8, 2021, <https://www.bloomberg.com/news/articles/2021-04-08/first-india-ev-battery-plant-takes-shape-to-cut-china-dependence>; "Sourcing Local, Supplying Global," Epsilon Advanced Materials, accessed February 29, 2024, <https://www.epsilonam.com/>.

For the production potential of anode material, see Epsilon Advanced Materials, "Epsilon Advanced Materials (EAM) Announces Investment of \$650M Manufacturing Facility in North Carolina to Strengthen EV Battery Industry in the United States," October 26, 2023, https://www.epsilonam.com/pdf/EAM-NC-Announcement-press-release_Final.pdf; Epsilon Advanced Materials, "Epsilon Advanced Materials (EAM) Announces Investment of \$650M Manufacturing Facility in North Carolina to Strengthen EV Battery Industry in the United States," October 26, 2023, https://www.epsilonam.com/pdf/EAM-NC-Announcement-press-release_Final.pdf; "Epsilon's Anode Manufacturing Plans and Recommendations to Support Component Manufacturing in India," *EVreporter*, October 11, 2023, <https://evreporter.com/epsilons-anode-manufacturing-plans-and-recommendation-to-support-component-manufacturing-in-india/>; Rishi Ranjan Kala, "TACC to Finalise Land for Anode Manufacturing Facility by December," *The Hindu BusinessLine*, December 7, 2022, <https://www.thehindubusinessline.com/companies/tacc-to-finalise-land-for-anode-manufacturing-facility-by-december/article66231019.ece>.

For the production potential of lithium, see Ministry of Mines, "Lithium Discovery in Jammu and Kashmir," July 31, 2023, <https://pib.gov.in/PressReleasePage.aspx?PRID=1944302>.

For the production potential of spherical graphite, see Jacqueline Holman, "Insight Conversation: Shishir Poddar, Tirupati Graphite," S&P Global, March 4, 2022, <https://www.spglobal.com/commodityinsights/en/market-insights/blogs/metals/030422-ic-tirupati-graphite-shishir-poddar>; Tirupati Graphite, "Successful Trials of Battery-Grade Spherical Graphite," press release, February 22, 2021, https://www.rns-pdf.londonstockexchange.com/rns/7618P_1-2021-2-19.pdf; Tirupati Graphite, "Sustainability Report 2021," 2021, 5, <https://tirupatigraphite.co.uk/images/Sustainability%20Report%2021.pdf>.

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INTERNATIONAL PARTNERSHIPS TO ADVANCE CLEAN ENERGY MANUFACTURING IN INDIA

BY SHAYAK SENGUPTA

1. INTRODUCTION

India has positioned itself as a growing producer of global clean energy technologies and their inputs, especially as countries like the United States look to turn away from China as a leading source of such technologies.¹ India's interest in taking advantage of geopolitical and geoeconomic fragmentation to capture economic opportunities from the clean energy transition is not unique among emerging markets and developing regions.² For example, Indonesia, Zambia, Namibia, and Chile have all announced efforts to capitalize on their mineral wealth (key to producing electric vehicle batteries) by either banning exports of raw materials to capture benefits from mineral processing or exerting greater public ownership of mineral assets.³

India's recent presidency of the Group of 20 (G20) advanced and emerging economies in 2023 highlighted the country's uniqueness among emerging economies. Not only did the presidency highlight the India's policy priorities in expanding global clean energy supply chains and associated finance, but the country was able to bridge geopolitical differences in these priorities across G20 nations.⁴ Moreover, the Indian G20 presidency's focus and outcomes on these topics elevated them for attention by G20 national governments.

The Indian G20 presidency bolsters the case for international partnerships to further integrate India into global supply chains and manufacturing of clean energy technologies. In addition, India's leadership in bodies like the International Solar Alliance (ISA) to deploy solar energy in other developing regions like Africa, show the country's rising clout in international climate policy.⁵ This, coupled with the staggering scale of the country's energy transition needs (in volumes of both clean energy and finance), and the country's attractiveness for renewable energy investment make India a crucial partner for any national or multilateral actor looking to advance the global energy transition and consequently, climate goals.⁶

However, beyond addressing global climate goals, these international partnerships have the potential to address numerous other needs: climate finance, economic development, and geopolitical objectives. Consequently, efforts to channel climate finance and technology to India and the international partnerships that underpin them must seize upon the opportunity to grow clean energy manufacturing in India.

¹ Dipka Bhambhani, "Biden Looks to India to Help Counter China Clean Energy Dominance," *Forbes*, December 11, 2023, <https://www.forbes.com/sites/dipkabhambhani/2023/12/11/biden-looks-to-india-to-help-counter-china-clean-energy-dominance/>.

² Charlotte Gardes-Landolfini, Pierpaolo Grippa, William Oman, and Sha Yu, "Staff Climate Notes: Energy Transition and Geoeconomic Fragmentation," International Monetary Fund, November 2023, 1, <https://doi.org/10.5089/9798400258220.066>.

³ Bentley Allan, Noah Gordon, and Jonas Goldman, "Building a Net-Zero World: How U.S. Finance Can Strengthen Clean Energy Manufacturing Abroad," Carnegie Endowment for International Peace, September 2023, 3, https://carnegieendowment.org/files/AGG_Building_Net_Zero_World_final3.pdf.

⁴ Indian G20 Presidency, "G20 New Delhi Leaders' Declaration," September 10, 2023, <https://www.mea.gov.in/Images/CPV/G20-New-Delhi-Leaders-Declaration.pdf>.

⁵ Vyoma Jha, "Politics: India's Political Leadership and Climate Change," in *The Making of the International Solar Alliance*, by Vyoma Jha, Oxford University Press Oxford, 2023, 101–130, <https://doi.org/10.1093/oso/9780198884705.003.0004>.

⁶ International Energy Agency, "India Energy Outlook 2021," *World Energy Outlook Special Report* (Paris, 2021), <https://www.iea.org/reports/india-energy-outlook-2021>; Maria Eugenia Mitri et al., "Climatescope 2023: Power Transition Factbook," BloombergNEF, November 29, 2023, <https://www.global-climatescope.org/downloads/climatescope-2023-report-en.pdf>.

2. OUTCOMES IN CLEAN ENERGY MANUFACTURING AND FINANCE FROM INDIA'S G20 PRESIDENCY

India's G20 presidency in 2023 elevated several priorities in climate change, energy transitions, and finance. Among them, G20 countries during India's presidency agreed to work to increase uptake of emerging climate and clean energy technologies, to diversify and grow clean energy manufacturing and supply chains, and to finance and grow investment to address climate change, especially from private sector sources.⁷ Two working groups of the G20 in 2023 focused on climate and clean energy technology, manufacturing, and finance during the Indian presidency: the Sustainable Finance Working Group (SFWG), led by finance ministers and central bank governors, and the Energy Transitions Working Group (ETWG), led by energy ministers.

Outcomes and recommendations from SFWG during India's G20 presidency adopted a paradigm of de-risking private investment for climate change, i.e. using strategic investments of public sector funding to reduce risk for private investors, thereby crowding-in private capital.⁸ This is especially relevant for developing regions which face bottlenecks from higher costs of capital due to real or perceived investment risks for private actors. Instruments to reduce these private sector investment risks include co-investments, insurance, or guarantees with public funding in the event of an investment project defaulting or failing. Coupled with these instruments include other recommendations for consistent regulatory environments and calls upon the multilateral development banks, such as the World Bank and Asian Development Bank, and Development Finance Institutions, like the U.S. Development Finance Corporation and the Japan Bank for International Cooperation, to increase the share of private sector investment in their investment projects.⁹

For emerging climate and clean energy technologies, SFWG continued its theme of de-risking private finance. It recommended national governments and multilateral institutions incentivize private sector research, development, and innovation investments in this area. They can do so by offering consistent regulatory environments, engage in public-private partnerships, and offering tools as loans, guarantees and tax incentives.¹⁰ While these recommendations focused on early-stage climate technologies, not commercial projects, innovation is key because many of the technologies needed to decarbonize have yet to reach commercial scale.¹¹ Moreover, to meet the desire of developing regions to capture the economic benefits of clean energy manufacturing, they must participate in higher-value technology development and innovation.¹²

⁷ Indian G20 Presidency, "G20 New Delhi Leaders' Declaration."

⁸ "2023 G20 Sustainable Finance Report Volume I," 2023, <https://g20sfwg.org/wp-content/uploads/2023/10/Volume-I-G20-India-Final-VF.pdf>.

⁹ "2023 G20 Sustainable Finance Report Volume I," 6–9.

¹⁰ "2023 G20 Sustainable Finance Report Volume I," 10–13.

¹¹ "2023 G20 Sustainable Finance Report Volume I," 10; Asian Development Bank, "Policy Measures and Financial Instruments for Catalysing the Rapid Development and Deployment of Green and Low-Carbon Technologies," Asian Development Bank, June 2023, <https://doi.org/10.1017/9781009157926.018>.

¹² Bentley Allan, Noah Gordon, and Jonas Goldman, "Building a Net-Zero World: How U.S. Finance Can Strengthen Clean Energy Manufacturing Abroad," 2–3.

The G20 Energy Ministers in 2023 through the ETWG agreed on several provisions related to supply chains of clean energy and low-cost financing for the energy transition. The ministers agreed to secure, diversified supply chains for critical minerals supporting the energy transition. G20 countries agreed to promote local value creation and finance to support technical and human resources in source regions, many of which are low and middle-income.¹³ Moreover, the ministers addressed technology gaps in the energy transition especially for developing countries. They agreed for the need for developing countries to access emerging clean energy technology through private and public channels as well as innovation, research, and development to deploy clean energy technologies in these regions.¹⁴ Lastly, like the finance ministers and central bank governors, G20 energy ministers agreed to support greater private sector financing for the energy transition through de-risking by public financing.¹⁵

3. THE NEED FOR STRONGER CLIMATE FINANCE AND TECHNOLOGY PARTNERSHIPS WITH INDIA

The Indian G20 presidency produced two overall themes related to the pursuit of expanding clean energy manufacturing:

- Greater integration of developing regions like India into global clean energy supply chains so they reap economic benefits.
- Increasing private sector investment in developing regions for this integration by strategic de-risking with public funds.

These themes complement India's own domestic efforts to grow local clean energy manufacturing. These include the country's Production-Linked Incentives (PLI), suites of financial and non-financial incentives aim to attract private sector investment for clean energy manufacturing. Relevant PLI schemes cover solar, battery and hydrogen equipment manufacturing.¹⁶

However, the fiscal latitude from the India's government is limited. The opportunity cost of a dollar in India is higher than wealthier economies. Likewise current financial flows for India's energy transition fall short of meeting the country's needs, with in-

¹³ G20 Energy Transitions Working Group, "G20 Energy Transitions Ministers' Meeting Outcome Document and Chair's Summary" (Goa, India, July 22, 2023), https://www.g20.in/content/dam/gtwenty/gtwenty_new/document/ETMM_ODCS.pdf; G20 Energy Transitions Working Group, "Voluntary High-Level Principles for Collaboration on Critical Minerals for Energy Transitions, Annex A," 2023, https://www.g20.in/content/dam/gtwenty/gtwenty_new/document/G20_ETWG_Annex_A.pdf.

¹⁴ G20 Energy Transitions Working Group, "G20 Energy Transitions Ministers' Meeting Outcome Document and Chair's Summary."

¹⁵ G20 Energy Transitions Working Group, "G20 Energy Transitions Ministers' Meeting Outcome Document and Chair's Summary,"; G20 Energy Transitions Working Group, "Voluntary Action Plan for Lowering the Cost of Finance for Energy Transitions, Annex C," 2023, https://www.g20.in/content/dam/gtwenty/gtwenty_new/document/G20_ETWG_Annex_C.pdf.

¹⁶ Uma Gupta, "21 Companies Bid for Electrolyzer Incentives in India," *Pv Magazine India*, December 19, 2023, <https://www.pv-magazine-india.com/2023/12/19/21-companies-bid-for-electrolyzer-incentives-in-india/>; Saurav Anand, "Seven Firms Bid for India's Advanced Chemistry Cell Manufacturing under PLI Scheme," *ET EnergyWorld*, April 24, 2024, <https://energy.economictimes.indiatimes.com/news/power/seven-firms-bid-for-indias-advanced-chemistry-cell-manufacturing-under-pli-scheme/109547871>; Monika Mondal and Mitul Kajaria, "India Wants to Build a Solar Manufacturing Ecosystem to Rival China," *Climate Home News*, January 30, 2024, <https://cleanenergyfrontier.climatechangenews.com/india-wants-own-solar-industry-break-reliance-china/>.

ternational flows a much lower fraction than domestic flows.¹⁷ Consequently, stronger international partnerships in clean energy manufacturing have the potential to not only deliver more climate finance for India's energy transition needs, but do so in a way that increase local value creation and seizes upon current geopolitical trends to diversify manufacturing away from China. These international partnerships could involve a variety of tools and action by the public and private sectors:

- Concessional finance by development finance institutions and governments to de-risk investment for private sector and complement PLI efforts.
- Governments encouraging private sector joint ventures and offtake agreements between Indian companies and companies abroad.
- Economic diplomacy and trade missions by governments to and from India to highlight possible business opportunities.
- Carve outs to qualify Indian component exports for financial incentives to encourage domestic manufacturing in export destinations like those of the Inflation Reduction Act.
- Governments continue highlighting the issue in bilateral and multilateral fora like G20.

In the production of green hydrogen, higher costs of capital for countries like India could cancel any advantage the country has from cheaper, more abundant solar resources. Concessional finance through debt, equity or guarantee instruments from multilateral and bilateral development finance institutions could lower this cost of capital. Likewise, participation of these public institutions gives confidence to private investors, thereby increasing chances of private sector finance. For example, several of these institutions have announced intentions to devote resources to hydrogen in emerging markets and developing countries, but India is among one of many geographies competing for this scarce public capital.¹⁸

An added advantage of India versus other emerging markets that makes it an attractive destination for international concessional finance is domestic demand for energy and associated products. Investment for projects to produce green hydrogen will only flow when there is a guaranteed offtake, or buyer, for the produced product, and the country is a large demand center. India is one of the largest producers of fertilizer and refined petroleum, two key sectors that use hydrogen.¹⁹ Consequently, offtake agreements for green hydrogen and its derivatives could create the demand signal needed for a growing

¹⁷ Neha Khanna, Dhruva Purkayastha, and Shreyans Jain, "Landscape of Green Finance in India: India's Green Investment Flows in FY 2019-20," Climate Policy Initiative, August 2022, <https://www.climatepolicyinitiative.org/wp-content/uploads/2022/08/Landscape-of-Green-Finance-in-India-2022-Full-Report.pdf>.

¹⁸ "International Energy Agency, "KfW Hydrogen Funds," IEA, February 1, 2023, <https://www.iea.org/policies/16995-kfw-hydrogen-funds>; Energy Sector Management Assistance Program, "Hydrogen for Development Partnership (H4D)," World Bank, March 2024, https://www.esmap.org/Hydrogen_for_Development_Partnership_H4D; Japan Bank for International Cooperation, "Aiming to Build a Global-Scale Hydrogen Supply Chain to Simultaneously Realize Carbon Neutrality, Economic Growth, and Energy Security," Japan Bank for International Cooperation, 2023, https://www.jbic.go.jp/en/information/today/today_202310/jtd_202310_pj1.html.

¹⁹ "Global Nitrogen Fertilizer Production by Country," Statista, May 23, 2023, <https://www.statista.com/statistics/1252656/nitrogen-fertilizer-production-by-country/>; "Refined Oil Products Production by Country," Statista, January 18, 2024, <https://www.statista.com/statistics/1445672/refined-oil-products-production-by-country-worldwide/>.

hydrogen market. For example, several Indian companies have already entered agreements to produce green hydrogen or ammonia for use by Japanese power and heavy industry. While these offtake agreements occur in the highly uncertain export market (as opposed to fulfilling domestic demand), they showcase international partnerships by governments to facilitate such agreements.²⁰ To meet domestic demand through offtake agreements, foreign direct investment catalyzed by concessional finance likewise could spur the green hydrogen market in India. These investments could also see the participation of state-owned enterprises controlled by the India's central government, which have sizable roles in fertilizer production and petroleum refining.²¹

In solar and battery manufacturing which offer more mature market than hydrogen, partnerships with India should target upstream, portions of their supply chains, many of which are capital-intensive. In the solar supply chain, raw polysilicon production along with ingot and wafer production have the highest barriers to entry to reorient away from China. Here, while India has lower initial investment costs, higher electricity costs make investments in these energy intensive segments of the solar supply chain difficult.²² Consequently, partnerships should finance renewable electricity and energy storage projects to provide firm, clean power for solar manufacturing. Likewise in the electric vehicle battery supply chain, India lacks significant reserve of key minerals required to manufacture batteries. While the country looks to secure supplies of these minerals through international partnerships like the U.S.-led Minerals Security Partnership and state-owned enterprises like Khanij Bidesh India Limited, international partnerships can again de-risk investment to create local value. Specifically, partnerships could focus on ancillary and precursor materials needed for battery production like graphite and manganese that India already produces as well as projects to recycle batteries.²³ Moreover, international partnerships could focus on public-private finance for capital-intensive battery cell manufacturing in India to complement attention by the PLI.²⁴

4. CONCLUSION

With greater geopolitical fragmentation, India has emerged as a growing manufacturer of global energy technologies and inputs. While unlikely to meet the scale of the current dominant player, China, India can be an alternative producer, especially for countries concerned by Chinese dominance. This is especially relevant as the country's

²⁰ "Indian Renewable Energy Company - ACME and Japanese Heavy Industry Major IHI Sign One of the Largest Pacts to Supply Green Ammonia from India to Japan," Press Information Bureau, January 23, 2024, <https://pib.gov.in/pib.gov.in/Pressreleaseshare.aspx?PRID=1998871>; "JERA Has Concluded an Agreement with ReNew to Jointly Develop a Green Ammonia Production Project" JERA, April 19, 2024, https://www.jera.co.jp/en/news/information/20240419_1889.

²¹ Department of Fertilizers, "Annual Report 2022-2023," Ministry of Chemicals and Fertilizers, Government of India, 2023, 135–36, <https://www.fert.nic.in/sites/default/files/2020-082024-01/Annual%20Report%20fertilizer%20new%20new17%20%20%2029-07-2023.pdf>; Ministry of Petroleum and Natural Gas, "Annual Report 2022-2023," Government of India, 2023, 271, <https://mopng.gov.in/files/TableManagements/annualreport2223.pdf>.

²² Gregory Wischer, "India: The Rising Power in Global Solar Photovoltaic Supply Chains," ORF America, May 2024, <https://orfamerica.org/newresearch/solar-rising-india>.

²³ Gregory Wischer, "Charging Up India's Potential Role in Global Battery Supply Chains," ORF America, May 2024, <https://orfamerica.org/newresearch/batteries-ev-india>.

²⁴ Saurav Anand, "Seven Firms Bid for India's Advanced Chemistry Cell Manufacturing under PLI Scheme."

presidency of the G20 in 2023 highlighted the issue and bridged geopolitical divides in it. India built consensus among the G20 to strategically de-risk private sector finance for climate action and create local value through reorienting clean energy supply chains. This supports the case for international partnerships that seek to further integrate India into global supply chains of clean energy technologies. These partnerships can extend the consensus at the G20 to address several priorities simultaneously: the need to channel more climate finance to India, India's economic aspirations and energy security needs, and geopolitical concerns about concentrated clean energy supply chains in China. International partnerships between the public and private sectors to increase clean energy manufacturing capacity in India can use financial and non-financial tools that increase finance and reduce risk for private sector investors.

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